

Numerical Methods in Computational Statistical Physics

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“Molecular Dynamics: Ab-Initio and Classical Methods”

Outline

- **Sampling the microcanonical ensemble** [Lectures 1-2]
 - Hamiltonian dynamics and ergodic assumption
 - Longtime numerical integration of the Hamiltonian dynamics
- **Sampling the canonical ensemble** [Lectures 2-3-4]
 - Stochastic differential equations (Langevin dynamics)
 - Markov chain approaches (Metropolis-Hastings)
 - Deterministic methods (Nosé-Hoover and the like)
- **Sampling other thermodynamic ensembles** [Lecture 4]
 - grand-canonical ensemble
 - isobaric-isothermal ensemble
- **Computation of free energy differences** [Lecture 5]

General references (1)

- Statistical physics: **theoretical** presentations
 - R. Balian, *From Microphysics to Macrophysics. Methods and Applications of Statistical Physics*, volume I - II (Springer, 2007).
 - many other books: Chandler, Ma, Phillies, Zwanzig, ...
- **Computational** Statistical Physics
 - D. Frenkel and B. Smit, *Understanding Molecular Simulation, From Algorithms to Applications* (Academic Press, 2002)
 - M. Tuckerman, *Statistical Mechanics: Theory and Molecular Simulation* (Oxford, 2010)
 - M. P. Allen and D. J. Tildesley, *Computer simulation of liquids* (Oxford University Press, 1987)
 - D. C. Rapaport, *The Art of Molecular Dynamics Simulations* (Cambridge University Press, 1995)
 - T. Schlick, *Molecular Modeling and Simulation* (Springer, 2002)

General references (2)

- Longtime integration of the **Hamiltonian** dynamics
 - E. Hairer, C. Lubich and G. Wanner, *Geometric Numerical Integration: Structure-Preserving Algorithms for ODEs* (Springer, 2006)
 - B. J. Leimkuhler and S. Reich, *Simulating Hamiltonian dynamics*, (Cambridge University Press, 2005)
 - E. Hairer, C. Lubich and G. Wanner, Geometric numerical integration illustrated by the Störmer-Verlet method, *Acta Numerica* **12** (2003) 399–450
- Sampling the **canonical** measure
 - L. Rey-Bellet, Ergodic properties of Markov processes, *Lecture Notes in Mathematics*, **1881** 1–39 (2006)
 - E. Cancès, F. Legoll and G. Stoltz, Theoretical and numerical comparison of some sampling methods, *Math. Model. Numer. Anal.* **41**(2) (2007) 351–390
 - T. Lelièvre, M. Rousset and G. Stoltz, *Free Energy Computations: A Mathematical Perspective* (Imperial College Press, 2010)
- J.N. Roux, S. Rodts and G. Stoltz, *Introduction à la physique statistique et à la physique quantique*, cours Ecole des Ponts (2009)
<http://cermics.enpc.fr/~stoltz/poly-phys-stat-quantique.pdf>

Microscopic description of physical systems: unknowns

- **Microstate** of a classical system of N particles:

$$(q, p) = (q_1, \dots, q_N, p_1, \dots, p_N) \in \mathcal{E}$$

Positions q (configuration), **momenta** p (to be thought of as $M\dot{q}$)

- In the simplest cases, $\mathcal{E} = \mathcal{D} \times \mathbb{R}^{3N}$ with $\mathcal{D} = \mathbb{R}^{3N}$ or $\mathbb{T}^{3N}$
- More complicated situations can be considered: molecular **constraints** defining submanifolds of the phase space
- **Hamiltonian** $H(q, p) = E_{\text{kin}}(p) + V(q)$, where the kinetic energy is

$$E_{\text{kin}}(p) = \frac{1}{2} p^T M^{-1} p, \quad M = \begin{pmatrix} m_1 \text{Id}_3 & & 0 \\ & \ddots & \\ 0 & & m_N \text{Id}_3 \end{pmatrix}.$$

Thermodynamic ensembles (1)

- **Macrostate** of the system described by a **probability measure**

Equilibrium thermodynamic properties (pressure,...)

$$\langle A \rangle_\mu = \mathbb{E}_\mu(A) = \int_{\mathcal{E}} A(q, p) \mu(dq dp)$$

- Choice of **thermodynamic ensemble**
 - **least biased** measure compatible with the observed **macroscopic** data
 - Volume, energy, number of particles, ... fixed **exactly or in average**
 - Equivalence of ensembles (as $N \rightarrow +\infty$)
- Constraints satisfied in average: constrained maximisation of entropy

$$S(\rho) = -k_B \int \rho \ln \rho d\lambda,$$

(λ reference measure), conditions $\rho \geq 0$, $\int \rho d\lambda = 1$, $\int A_i \rho d\lambda = \mathcal{A}_i$

An example: the NVT ensemble

- **Canonical** ensemble: reference measure $d\lambda =$ uniform measure on (q, p)
- Constraint: **average energy** fixed $A_0 = H$
- Euler-Lagrange equations

$$S'(\rho) + \alpha + \beta H = 0$$

Lagrange multipliers α, β respectively related to the constraints

$$\int_{\mathcal{E}} \rho(q, p) dq dp = 1, \quad \int_{\mathcal{E}} H(q, p) \rho(q, p) dq dp = E_0$$

Compute $S'(\rho) = 1 + \ln \rho$, so that $\ln \rho = 1 - \alpha - \beta H$

Canonical measure

$$\mu_{\text{NVT}}(dq dp) = Z_{\text{NVT}}^{-1} e^{-\beta H(q, p)} dq dp$$

Define the **temperature** as $T = \frac{1}{k_B \beta}$

Observables

- May **depend on the chosen ensemble!** Given by physicists, by some **analogy** with macroscopic, continuum thermodynamics

- Pressure (derivative of the free energy with respect to volume)

$$A(q, p) = \frac{1}{3|\mathcal{D}|} \sum_{i=1}^N \left(\frac{p_i^2}{m_i} - q_i \cdot \nabla_{q_i} V(q) \right)$$

- Kinetic temperature $A(q, p) = \frac{1}{3Nk_B} \sum_{i=1}^N \frac{p_i^2}{m_i}$

- Specific heat at constant volume: **canonical** average

$$C_V = \frac{\mathcal{N}_a}{Nk_B T^2} \left(\langle H^2 \rangle_{\text{NVT}} - \langle H \rangle_{\text{NVT}}^2 \right)$$

Main issue

Computation of **high-dimensional** integrals... **Ergodic** averages

- Also techniques to compute **transport coefficients** (not presented here)

Sampling the microcanonical ensemble

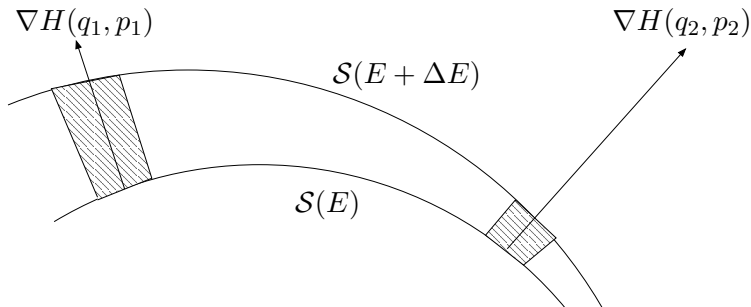
- **Sampling the microcanonical measure**
 - Definition of the microcanonical measure
 - The Hamiltonian dynamics and its properties
 - The ergodic assumption
- **Standard numerical analysis of ordinary differential equations**
 - Consistency, stability, convergence
 - Standard examples
- **Longtime numerical integration of the Hamiltonian dynamics**
 - Failure of standard schemes
 - Symplecticity and construction of symplectic schemes
 - Elements of backward error analysis

The microcanonical measure

Lebesgue measure conditioned to $\mathcal{S}(E) = \left\{ (q, p) \in \mathcal{E} \mid H(q, p) = E \right\}$
(co-area formula)

Microcanonical measure

$$\mu_{\text{mc},E}(dq dp) = Z_E^{-1} \delta_{H(q,p)-E}(dq dp) = Z_E^{-1} \frac{\sigma_{\mathcal{S}(E)}(dq dp)}{|\nabla H(q, p)|}$$



The Hamiltonian dynamics (1)

Hamiltonian dynamics

$$\begin{cases} \frac{dq(t)}{dt} = \nabla_p H(q(t), p(t)) = M^{-1}p(t) \\ \frac{dp(t)}{dt} = -\nabla_q H(q(t), p(t)) = -\nabla V(q(t)) \end{cases}$$

Assumed to be well-posed (e.g. when the energy is a Lyapunov function)

- **Flow**: $\phi_t(q_0, p_0)$ solution at time t starting from initial condition (q_0, p_0)
- Why Hamiltonian formalism? (instead of working with velocities?)
 - Note that the vector field is divergence-free

$$\operatorname{div}_q \left(\nabla_p H(q(t), p(t)) \right) + \operatorname{div}_p \left(-\nabla_q H(q(t), p(t)) \right) = 0$$

- **Volume** preservation $\int_{\phi_t(B)} dq dp = \int_B dq dp$

The Hamiltonian dynamics (2)

- Other properties

- Preservation of **energy** $H \circ \phi_t = H$

$$\frac{d}{dt} \left[H(q(t), p(t)) \right] = \nabla_q H(q(t), p(t)) \cdot \frac{dq(t)}{dt} + \nabla_p H(q(t), p(t)) \cdot \frac{dp(t)}{dt} = 0$$

- **Time-reversibility** $\phi_{-t} = S \circ \phi_t \circ S$ where $S(q, p) = (q, -p)$

Proof: use $S^2 = \text{Id}$ and note that

$$S \circ \phi_{-t}(q_0, p_0) = (q(-t), -p(-t))$$

is a solution of the Hamiltonian dynamics starting from $(q_0, -p_0)$, as is $\phi_t \circ S(q_0, p_0)$. Conclude by uniqueness of solution.

- **Symmetry** $\phi_{-t} = \phi_t^{-1}$ (in general, $\phi_{t+s} = \phi_t \circ \phi_s$)

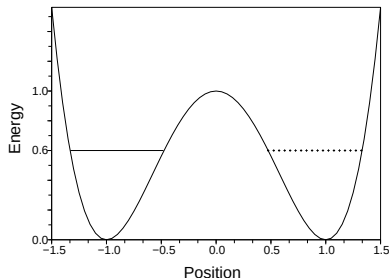
Ergodicity of the Hamiltonian dynamics

- Invariance of the microcanonical measure by the Hamiltonian dynamics

Ergodic assumption

$$\langle A \rangle_{\text{NVE}} = \int_{S(E)} A(q, p) \mu_{\text{mc}, E}(dq dp) = \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T A(\phi_t(q, p)) dt$$

- Wrong when **spurious invariants** are conserved, such as $\sum_{i=1}^N p_i$



Numerical approximation

- The ergodic assumption is true...
 - for **completely integrable** systems and perturbations thereof (KAM), upon **conditioning** the microcanonical measure by all invariants
 - if **stochastic perturbations** are considered¹
- Although questionable, ergodic averages are the only **realistic** option
- Requires trajectories with **good energy preservation** over **very long times**
→ **disqualifies default schemes** (Explicit/Implicit Euler, RK4, ...)
- Standard (simplest) estimator: integrator $(q^{n+1}, p^{n+1}) = \Phi_{\Delta t}(q^n, p^n)$

$$\langle A \rangle_{\text{NVE}} \simeq \frac{1}{N_{\text{iter}}} \sum_{n=1}^{N_{\text{iter}}} A(q^n, p^n)$$

or refined estimators using some filtering strategy²

¹E. Faou and T. Lelièvre, *Math. Comput.* **78**, 2047–2074 (2009)

²Cancès *et. al*, *J. Chem. Phys.*, 2004 and *Numer. Math.*, 2005

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Some fundamentals of numerical integration of ODEs

- Consider an **ordinary differential equation** $\frac{dy(t)}{dt} = f(y(t))$
- Assume that it is well posed (unique solution for all initial conditions)

$$y(t) = \phi_t(y(0)) = y(0) + \int_0^t f(y(s)) ds$$

- Introduce y^n , approximation of $y(t_n)$ with $t_n = n\Delta t$ (fixed time step)

One step method

$$y^{n+1} = \Phi_{\Delta t}(y^n)$$

- Simplest example: Explicit Euler

$$y^{n+1} = y^n + \Delta t f(y^n)$$

in which case $\Phi_{\Delta t}(y) = y + \Delta t f(y)$

Further examples

- **Explicit** methods

- Heun: $y^{n+1} = y^n + \frac{\Delta t}{2} \left(f(y^n) + f(y^n + \Delta t f(y^n)) \right)$

- Fourth order Runge-Kutta scheme

$$y^{n+1} = y^n + \Delta t \frac{f(y^n) + 2f(Y^{n+1}) + 2f(Y^{n+2}) + f(Y^{n+3})}{6}$$

with $Y^{n+1} = y^n + f(Y^n) \frac{\Delta t}{2}$, $Y^{n+2} = y^n + f(Y^{n+1}) \frac{\Delta t}{2}$, and
 $Y^{n+3} = y^n + f(Y^{n+2}) \Delta t$

- **Implicit** methods [solve using a fixed-point iteration for instance]

- Implicit Euler: $y^{n+1} = y^n + \Delta t f(y^{n+1})$

- Trapezoidal rule: $y^{n+1} = y^n + \frac{\Delta t}{2} \left(f(y^n) + f(y^{n+1}) \right)$

- Midpoint: $y^{n+1} = y^n + \Delta t f \left(\frac{y^n + y^{n+1}}{2} \right)$

Standard error analysis

- Error on the **trajectory over finite times**
 - **local** error at each time step (consistency + rounding off error)
 - **accumulation** of the errors (stability)
- A numerical method is **convergent** when the **global** error satisfies

$$\lim_{\Delta t \rightarrow 0} \left(\max_{0 \leq n \leq N} \|y^n - y(n\Delta t)\| \right) = 0$$

- **Order p consistency**: quantification of the error over **one time step**

$$e(y_0) = y(\Delta t) - y_0 - \Delta t \Phi(y_0) = O(\Delta t^{p+1})$$

- Example: explicit Euler is of order 1 $\rightarrow$ Taylor expansion

$$y(\Delta t) - \left(y_0 + \Delta t f(y_0) \right) = \frac{\Delta t^2}{2} y''(\theta \Delta t), \quad y''(\tau) = \partial_y f(y(\tau)) \cdot f(y(\tau))$$

Standard error analysis

- **Stability**: for all sequences $y^{n+1} = \Phi_{\Delta t}(y^n)$ and $z^{n+1} = \Phi_{\Delta t}(z^n) + \delta^n$, it holds (S independent of Δt)

$$\max_{0 \leq n \leq N} \|y^n - z^n\| \leq S \left(|y^0 - z^0| + \sum_{n=0}^N \|\delta^n\| \right)$$

True when $\|\Phi_{\Delta t}(y_1) - \Phi_{\Delta t}(y_2)\| \leq \Lambda \|y_1 - y_2\|$

- A method which is **stable and consistent** is **convergent**
(take $z^n = y(n\Delta t)$ exact solution, so that δ_n is the local truncation error)
- For a method of order p , there are $N = [T/\Delta t]$ integration steps

$$\max_{0 \leq n \leq N} \|y^n - y(t_n)\| \leq C(T) \Delta t^p$$

with a prefator which typically **grows exponentially with T** ...

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Longtime integration: failure of default schemes

- Appropriate notion of **stability**: longtime **energy preservation**

Hamiltonian dynamics as a first-order differential equation

$$y = \begin{pmatrix} q \\ p \end{pmatrix}, \quad \dot{y} = J \nabla H(y), \quad J = \begin{pmatrix} 0 & I_{3N} \\ -I_{3N} & 0 \end{pmatrix}$$

- **Analytical study** of $\Phi_{\Delta t}$ for 1D **harmonic** potential $V(q) = \frac{1}{2}\omega^2 q^2$

$$\begin{cases} q^{n+1} = q^n + \Delta t M^{-1} p^n, \\ p^{n+1} = p^n - \Delta t \nabla V(q^n), \end{cases} \quad \text{so that } y^{n+1} = \begin{pmatrix} 1 & \Delta t \\ -\omega^2 \Delta t & 1 \end{pmatrix} y^n$$

Modulus of eigenvalues $|\lambda_{\pm}| = \sqrt{1 + \omega^2 \Delta t^2} > 1$, hence exponential **increase** of the energy

- For implicit Euler and Runge-Kutta 4 (for Δt small enough), exponential **decrease** of the energy
- **Numerical confirmation** for general (**anharmonic**) potentials

Which qualitative properties are important?

- **Time reversibility** $\Phi_{\Delta t} \circ S = S \circ \Phi_{-\Delta t}$ usually verified

Check it for Explicit Euler $\Phi_{\Delta t}^{\text{Euler}}(q, p) = (q + \Delta t M^{-1} p, p - \Delta t \nabla V(q))$

$$\Phi_{\Delta t}^{\text{Euler}}(q, -p) = \begin{pmatrix} q - \Delta t M^{-1} p \\ -p - \Delta t \nabla V(q) \end{pmatrix} = S \begin{pmatrix} q - \Delta t M^{-1} p \\ p + \Delta t \nabla V(q) \end{pmatrix} = S \left(\Phi_{-\Delta t}^{\text{Euler}}(q, p) \right)$$

- **Symmetry** $\Phi_{\Delta t}^{-1} = \Phi_{-\Delta t}$ is not trivial at all
- **Oriented volume preservation**: linear case in 2D
 - two independent vectors $q = (x, y)$ and $q' = (x', y')$, oriented volume

$$q \wedge q' = xy' - xy = q^T J q', \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

- linear transformation A , so that $q \rightarrow Aq$ and $q' \rightarrow Aq'$

$$q^T J q' \rightarrow q^T A^T J A q'$$

- unchanged provided $A^T J A = J$

Longtime integration: symplecticity (1)

- Generalization to higher dimensions and nonlinear transformations
 - mapping $g(q, p) = (g_1(q, p), \dots, g_{6N}(q, p))^T$
 - Jacobian matrix $g'(q, p)$

$$g'(q, p) = \begin{pmatrix} \frac{\partial g_1}{\partial q_1} & \cdots & \frac{\partial g_1}{\partial q_{3N}} & \frac{\partial g_1}{\partial p_1} & \cdots & \frac{\partial g_1}{\partial p_{3N}} \\ & \ddots & & & \ddots & \\ \frac{\partial g_{6N}}{\partial q_1} & \cdots & \frac{\partial g_{6N}}{\partial q_{3N}} & \frac{\partial g_{6N}}{\partial p_1} & \cdots & \frac{\partial g_{6N}}{\partial p_{2dN}} \end{pmatrix}.$$

Symplectic mapping

$$[g'(q, p)]^T J g'(q, p) = J$$

- A mapping is symplectic if and only if it is (locally) Hamiltonian
- A composition of symplectic mappings is symplectic

Longtime integration: symplecticity (2)

- Proof: A Hamiltonian mapping is symplectic

Derive the Jacobian matrix $\psi(t, y) = \frac{\partial \phi_t(y)}{\partial y}$

$$\frac{d\psi}{dt} = \frac{\partial}{\partial y} \left(\frac{d\phi_t(y)}{dt} \right) = \frac{\partial}{\partial y} (J \nabla H(\phi_t(y))) = J (\nabla^2 H(\phi_t(y))) \frac{\partial \phi_t(y)}{\partial y}$$

so that, using $J^T = -J$

$$\frac{d}{dt} \left(\psi(t)^T J \psi(t) \right) = \psi(t)^T (\nabla^2 H(\phi_t(y))) J^T J \psi(t) + \psi(t)^T (\nabla^2 H(\phi_t(y))) J^2 \psi(t) = 0$$

The conclusion follows since $\psi(0)^T J \psi(0) = J$. Converse statement: “integrability Lemma” (see Hairer/Lubich/Wanner, Theorem VI.2.6 and Lemma VI.2.7)

- Composition of symplectic mappings g, h : use $(g \circ h)' = (g' \circ h)h'$ and

$$h'(q, p)^T \left(g'(h(q, p)) \right)^T J \left(g'(h(q, p)) \right) h'(q, p) = [h'(q, p)]^T J h'(q, p) = J$$

Longtime integration: symplecticity (3)

- Stability result

Approximate longtime energy conservation

For an analytic Hamiltonian H and a symplectic method $\Phi_{\Delta t}$ of order p , and if the numerical trajectory remains in a compact subset, then there exists $h > 0$ and $\Delta t^* > 0$ such that, for $\Delta t \leq \Delta t^*$,

$$H(q^n, p^n) = H(q^0, p^0) + O(\Delta t^p)$$

for exponentially long times $n\Delta t \leq e^{h/\Delta t}$.

- Weaker results under weaker assumptions³
- Does not say anything on the **statistical behavior**! Near energy preservation is a **necessary** condition

³Hairer/Lubich/Wanner, 2006 and *Acta Numerica*, 2003

Longtime integration: constructing symplectic schemes (1)

- **Splitting** strategy for a general ODE $\dot{y}(t) = f(y)$, flow ϕ_t
 - Decompose the vector field as $f(y) = f_1(y) + f_2(y)$
 - Define the flows ϕ_t^i associated with each elementary ODE $\dot{z}(t) = f_i(z)$
 - Motivation: (almost) **analytical integration** of elementary ODEs
 - Generalization to a decomposition into $m \geq 2$ parts

- **Trotter** splitting (first order accurate)

$$\phi_{\Delta t} = \phi_{\Delta t}^1 \circ \phi_{\Delta t}^2 + O(\Delta t^2) = \phi_{\Delta t}^2 \circ \phi_{\Delta t}^1 + O(\Delta t^2)$$

- **Strang** splitting (second order)

$$\phi_{\Delta t} = \phi_{\Delta t/2}^1 \circ \phi_{\Delta t}^2 \circ \phi_{\Delta t/2}^1 + O(\Delta t^3) = \phi_{\Delta t/2}^2 \circ \phi_{\Delta t}^1 \circ \phi_{\Delta t/2}^2 + O(\Delta t^3)$$

- Extension to higher order schemes (Suzuki, Yoshida)

Longtime integration: constructing symplectic schemes (2)

- **Splitting** Hamiltonian systems: $\begin{cases} \dot{q} = M^{-1}p \\ \dot{p} = 0 \end{cases}$ and $\begin{cases} \dot{q} = 0 \\ \dot{p} = -\nabla V(q) \end{cases}$
- Flows $\phi_t^1(q, p) = (q + t M^{-1}p, p)$ and $\phi_t^2(q, p) = (q, p - t\nabla V(q))$
- **Symplectic Euler A**: first order scheme $\Phi_{\Delta t} = \phi_{\Delta t}^2 \circ \phi_{\Delta t}^1$

$$\begin{cases} q^{n+1} = q^n + \Delta t M^{-1} p^n \\ p^{n+1} = p^n - \Delta t \nabla V(q^{n+1}) \end{cases}$$

Composition of Hamiltonian flows hence symplectic

- Linear stability: harmonic potential $A(\Delta t) = \begin{pmatrix} 1 & \Delta t \\ -\omega^2 \Delta t & 1 - (\omega \Delta t)^2 \end{pmatrix}$
- Eigenvalues $|\lambda_{\pm}| = 1$ provided $\omega \Delta t < 2$
→ time-step limited by the highest frequencies

Longtime integration: symmetrization of schemes⁴

- **Strang splitting** $\Phi_{\Delta t} = \phi_{\Delta t/2}^2 \circ \phi_{\Delta t}^1 \circ \phi_{\Delta t/2}^2$, second order scheme

Störmer-Verlet scheme

$$\begin{cases} p^{n+1/2} = p^n - \frac{\Delta t}{2} \nabla V(q^n) \\ q^{n+1} = q^n + \Delta t M^{-1} p^{n+1/2} \\ p^{n+1} = p^{n+1/2} - \frac{\Delta t}{2} \nabla V(q^{n+1}) \end{cases}$$

- Properties:
 - Symplectic, symmetric, time-reversible
 - One force evaluation per time-step, linear stability condition $\omega \Delta t < 2$
 - In fact, $M \frac{q^{n+1} - 2q^n + q^{n-1}}{\Delta t^2} = -\nabla V(q^n)$

⁴L. Verlet, *Phys. Rev.* **159**(1) (1967) 98-105

Molecular constraints

- In some cases, mechanical systems are **constrained**
- Numerical motivation: **highly oscillatory** systems
 - Fast oscillations of the system, e.g. vibrations of bonds and bond angles
 - Severe limitations on admissible time steps since $\omega\Delta t < 2$
 - Remove the limitation by constraining these degrees of freedom
 - Introduces some sampling errors, which can be corrected
- Other motivation: computation of free energy difference with thermodynamic integration
- The Hamiltonian dynamics has to be modified consistently, and appropriate numerical schemes have to be devised (RATTLE)
→ more on Friday!

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Some elements of backward error analysis

- Philosophy of backward analysis for EDOs: the numerical solution is...
 - an **approximate solution of the exact dynamics** $\dot{y} = f(y)$
 - the **exact solution of a modified dynamics** : $y^n = z(t_n)$
- properties of numerical scheme deduced from properties of $\dot{z} = f_{\Delta t}(z)$

Modified dynamics

$$\dot{z} = f_{\Delta t}(z) = f(z) + \Delta t F_1(z) + \Delta t^2 F_2(z) + \dots, \quad z(0) = y^0$$

- For Hamiltonian systems ($f(y) = J\nabla H(y)$) **and** symplectic scheme:
*Exact conservation of an **approximate Hamiltonian** $H_{\Delta t}$, hence
approximate conservation of the exact Hamiltonian*
- Harmonic oscillator: $H_{\Delta t}(q, p) = H(q, p) - \frac{(\omega\Delta t)^2 q^2}{4}$ for Verlet

General construction of the modified dynamics

- **Iterative procedure** (carried out up to an arbitrary truncation order)
- Taylor expansion of the solution of the modified dynamics

$$z(\Delta t) = z(0) + \Delta t \dot{z}(0) + \frac{\Delta t^2}{2} \ddot{z}(0) + \dots$$

with $\begin{cases} \dot{z}(0) = f(z(0)) + \Delta t F_1(z(0)) + O(\Delta t^2) \\ \ddot{z}(0) = \partial_z f(z(0)) \cdot f(z(0)) + O(\Delta t) \end{cases}$

Modified dynamics: first order correction

$$z(\Delta t) = y^0 + \Delta t f(y^0) + \Delta t^2 \left(F_1(y^0) + \frac{1}{2} \partial_z f(y^0) f(y^0) \right) + O(\Delta t^3)$$

- To be **compared** to $y^1 = \Phi_{\Delta t}(y^0) = y^0 + \Delta t f(y^0) + \dots$

Some examples

- **Explicit Euler** $y^1 = y^0 + \Delta t f(y^0)$: the correction is **not Hamiltonian**

$$F_1(z) = -\frac{1}{2}\partial_z f(z)f(z) = \frac{1}{2} \begin{pmatrix} M^{-1}\nabla_q V(q) \\ \nabla_q^2 V(q) \cdot M^{-1}p \end{pmatrix} \neq \begin{pmatrix} \nabla_p H_1 \\ -\nabla_q H_1 \end{pmatrix}$$

- **Symplectic Euler A**

$$\begin{cases} q^{n+1} = q^n + \Delta t M^{-1} p^n, \\ p^{n+1} = p^n - \Delta t \nabla_q V(q^n) - \Delta t \nabla_q^2 V(q^n) M^{-1} p^n + O(\Delta t^3) \end{cases}$$

The correction derives from the **Hamiltonian** $H_1(q, p) = \frac{1}{2} p^T M^{-1} \nabla_q V(q)$

$$F_1(q, p) = \frac{1}{2} \begin{pmatrix} M^{-1}\nabla_q V(q) \\ -\nabla_q^2 V(q) \cdot M^{-1}p \end{pmatrix} = \begin{pmatrix} \nabla_p H_1(q, p) \\ -\nabla_q H_1(q, p) \end{pmatrix}$$

Energy $H + \Delta t H_1$ preserved at order 2, while H preserved only at order 1

Sampling the canonical ensemble

Classification of the methods

- Computation of $\langle A \rangle = \int_{\mathcal{E}} A(q, p) \mu(dq dp)$ with

$$\mu(dq dp) = Z_{\mu}^{-1} e^{-\beta H(q, p)} dq dp, \quad \beta = \frac{1}{k_B T}$$

- Actual issue: sampling canonical measure on configurational space

$$\nu(dq) = Z_{\nu}^{-1} e^{-\beta V(q)} dq$$

- Several strategies (theoretical and numerical comparison⁵)
 - **Purely stochastic** methods (i.i.d sample) $\rightarrow$ impossible...
 - **Stochastic differential equations**
 - **Markov chain** methods
 - **Deterministic methods** *à la* Nosé-Hoover

In practice, no clear-cut distinction due to **blending**...

⁵E. Cancès, F. Legoll and G. Stoltz, *M2AN*, 2007

- **Stochastic differential equations**
 - General perspective (convergence results, ...)
 - Overdamped Langevin dynamics (Einstein-Schmolukowski)
 - Langevin dynamics
 - Extensions: DPD, Generalized Langevin
- **Markov chain methods**
 - Metropolis-Hastings algorithm
- **Deterministic methods**
 - Nosé-Hoover and the like
 - Nosé-Hoover Langevin

Langevin dynamics

- **Stochastic** perturbation of the Hamiltonian dynamics : friction $\gamma > 0$

$$\begin{cases} dq_t = M^{-1} p_t dt \\ dp_t = -\nabla V(q_t) dt - \gamma M^{-1} p_t dt + \sqrt{\frac{2\gamma}{\beta}} dW_t \end{cases}$$

- Motivations
 - **Ergodicity** can be proved and is indeed observed in practice
 - Many **useful extensions** (dissipative particle dynamics, rigorous NPT and μ VT samplings, etc)
- Aims
 - Understand the **meaning** of this equation
 - Understand why it samples the canonical ensemble
 - Implement appropriate discretization schemes
 - Estimate the **errors** (systematic biases vs. statistical uncertainty)

An intuitive view of the Brownian motion (1)

- **Independant Gaussian increments** whose variance is proportional to time

$$\forall 0 < t_0 \leq t_1 \leq \dots \leq t_n, \quad W_{t_{i+1}} - W_{t_i} \sim \mathcal{N}(0, t_{i+1} - t_i)$$

where the increments $W_{t_{i+1}} - W_{t_i}$ are **independent**

- $G \sim \mathcal{N}(m, \sigma^2)$ distributed according to the probability density

$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

- The solution of $dq_t = \sigma dW_t$ can be thought of as the limit $\Delta t \rightarrow 0$

$$q^{n+1} = q^n + \sigma\sqrt{\Delta t} G^n, \quad G^n \sim \mathcal{N}(0, 1) \text{ i.i.d.}$$

where q^n is an approximation of $q_{n\Delta t}$

- Note that $q^n \sim \mathcal{N}(q^0, \sigma n \Delta t)$
- Multidimensional case: $W_t = (W_{1,t}, \dots, W_{d,t})$ where W_i are independent

An intuitive view of the Brownian motion (2)

- Analytical study of the process: **law** $\psi(t, q)$ of the process at time t
→ distribution of all possible realizations of q_t for
 - a given initial distribution $\psi(0, q)$, e.g. δ_{q^0}
 - and all realizations of the Brownian motion

Averages at time t

$$\mathbb{E}\left(A(q_t)\right) = \int_{\mathcal{D}} A(q) \psi(t, q) dq$$

- Partial differential equation governing the evolution of the law

Fokker-Planck equation

$$\partial_t \psi = \frac{\sigma^2}{2} \Delta \psi$$

Here, simple heat equation → **“diffusive behavior”**

An intuitive view of the Brownian motion (3)

- Proof: Taylor expansion, beware random terms of order $\sqrt{\Delta t}$

$$\begin{aligned} A(q^{n+1}) &= A\left(q^n + \sigma\sqrt{\Delta t} G^n\right) \\ &= A(q^n) + \sigma\sqrt{\Delta t} G^n \cdot \nabla A(q^n) + \frac{\sigma^2 \Delta t}{2} (G^n)^T (\nabla^2 A(q^n)) G^n + O(\Delta t^{3/2}) \end{aligned}$$

Taking expectations (Gaussian increments G^n independent from the current position q^n)

$$\mathbb{E}[A(q^{n+1})] = \mathbb{E}\left[A(q^n) + \frac{\sigma^2 \Delta t}{2} \Delta A(q^n)\right] + O(\Delta t^{3/2})$$

Therefore, $\mathbb{E}\left[\frac{A(q^{n+1}) - A(q^n)}{\Delta t} - \frac{\sigma^2}{2} \Delta A(q^n)\right] \rightarrow 0$. On the other hand,

$$\mathbb{E}\left[\frac{A(q^{n+1}) - A(q^n)}{\Delta t}\right] \rightarrow \partial_t (\mathbb{E}[A(q_t)]) = \int_{\mathcal{D}} A(q) \partial_t \psi(t, q) dq.$$

This leads to

$$0 = \int_{\mathcal{D}} A(q) \partial_t \psi(t, q) dq - \frac{\sigma^2}{2} \int_{\mathcal{D}} \Delta A(q) \psi(t, q) dq = \int_{\mathcal{D}} A(q) \left(\partial_t \psi(t, q) - \frac{\sigma^2}{2} \Delta \psi(t, q) \right) dq$$

This equality holds for all observables A .

General SDEs (1)

- State of the system $X \in \mathbb{R}^d$, m -dimensional Brownian motion, diffusion matrix $\sigma \in \mathbb{R}^{d \times m}$

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t$$

to be thought of as the limit as $\Delta t \rightarrow 0$ of (X^n approximation of $X_{n\Delta t}$)

$$X^{n+1} = X^n + \Delta t b(X^n) + \sqrt{\Delta t} \sigma(X^n) G^n, \quad G^n \sim \mathcal{N}(0, \text{Id}_m)$$

- Generator

$$\mathcal{L} = b(x) \cdot \nabla + \frac{1}{2} \sigma \sigma^T(x) : \nabla^2 = \sum_{i=1}^d b_i(x) \partial_{x_i} + \frac{1}{2} \sum_{i,j=1}^d [\sigma \sigma^T(x)]_{i,j} \partial_{x_i} \partial_{x_j}$$

- Proceeding as before, it can be shown that

$$\partial_t \left(\mathbb{E}[A(q_t)] \right) = \int_{\mathcal{X}} A \partial_t \psi = \mathbb{E} \left[(\mathcal{L}A)(X_t) \right] = \int_{\mathcal{X}} (\mathcal{L}A) \psi$$

General SDEs (2)

Fokker-Planck equation

$$\partial_t \psi = \mathcal{L}^* \psi$$

where $\mathcal{L}^*$ is the adjoint of $\mathcal{L}$

$$\int_{\mathcal{X}} (\mathcal{L}A)(x) B(x) dx = \int_{\mathcal{X}} A(x) (\mathcal{L}^*B)(x) dx$$

- Invariant measures are **stationary** solutions of the Fokker-Planck equation

Invariant probability measure $\psi_{\infty}(x) dx$

$$\mathcal{L}^* \psi_{\infty} = 0, \quad \int_{\mathcal{X}} \psi_{\infty}(x) dx = 1, \quad \psi_{\infty} \geq 0$$

- When $\mathcal{L}$ is elliptic (i.e. $\sigma\sigma^T$ has full rank: the **noise is sufficiently rich**), the process can be shown to be **irreducible** = accessibility property

$$P_t(x, \mathcal{S}) = \mathbb{P}(X_t \in \mathcal{S} \mid X_0 = x) > 0$$

General SDEs (3)

- Sufficient conditions for ergodicity
 - irreducibility
 - **existence** of an invariant probability measure $\psi_\infty(x) dx$

Then the invariant measure is **unique** and

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \varphi(X_t) dt = \int_{\mathcal{X}} \varphi(x) \psi_\infty(x) dx \quad \text{a.s.}$$

- Rate of convergence given by **Central Limit Theorem**: $\tilde{\varphi} = \varphi - \int \varphi \psi_\infty$

$$\sqrt{T} \left(\frac{1}{T} \int_0^T \varphi(X_t) dt - \int \varphi \psi_\infty \right) \xrightarrow[T \rightarrow +\infty]{\text{law}} \mathcal{N}(0, \sigma_\varphi^2)$$

with $\sigma_\varphi^2 = 2 \mathbb{E} \left[\int_0^{+\infty} \tilde{\varphi}(X_t) \tilde{\varphi}(X_0) dt \right]$ (proof: later, discrete time setting)

SDEs: numerics (1)

- Numerical discretization: various schemes (**Markov chains** in all cases)
- Example: Euler-Maruyama

$$X^{n+1} = X^n + \Delta t b(X^n) + \sqrt{\Delta t} \sigma(X^n) G^n, \quad G^n \sim \mathcal{N}(0, \text{Id}_d)$$

- Standard notions of error: **fixed integration time** $T < +\infty$
 - **Strong error** $\sup_{0 \leq n \leq T/\Delta t} \mathbb{E} |X^n - X_{n\Delta t}| \leq C \Delta t^p$
 - **Weak error** $\sup_{0 \leq n \leq T/\Delta t} \left| \mathbb{E} [\varphi(X^n)] - \mathbb{E} [\varphi(X_{n\Delta t})] \right| \leq C \Delta t^p$
 - “mean error” vs. “error of the mean”
- Example: for Euler-Maruyama, weak order 1, strong order 1/2 (1 when σ constant)

Generating (pseudo) random numbers (1)

- The basis is the generation of numbers uniformly distributed in $[0, 1]$
- **Deterministic** sequences which **look like** they are random...
 - Early methods: linear congruential generators (“chaotic” sequences)

$$x_{n+1} = ax_n + b \bmod c, \quad u_n = \frac{x_n}{c-1}$$

- Known defects: short periods, point alignments, etc, which can be (partially) patched by cleverly combining several generators
- More recent algorithms: shift registers, such as **Mersenne-Twister**
→ default choice in e.g. Scilab, available in the GNU Scientific Library
- **Randomness tests**: various flavors

Generating (pseudo) random numbers (2)

- Standard distributions are obtained from the uniform distribution by...

- inversion of the cumulative function** $F(x) = \int_{-\infty}^x f(y) dy$ (which is an increasing function from $\mathbb{R}$ to $[0, 1]$)

$$X = F^{-1}(U) \sim f(x) dx$$

Proof: $\mathbb{P}\{a < X \leq b\} = \mathbb{P}\{a < F^{-1}(U) \leq b\} = \mathbb{P}\{F(a) < U \leq F(b)\} = F(b) - F(a) = \int_a^b f(x) dx$

Example: exponential law of density $\lambda e^{-\lambda x} \mathbf{1}_{\{x \geq 0\}}$, $F(x) = \mathbf{1}_{\{x \geq 0\}}(1 - e^{-\lambda x})$, so that $X = -\frac{1}{\lambda} \ln U$

- change of variables:** standard Gaussian $G = \sqrt{-2 \ln U_1} \cos(2\pi U_2)$

Proof: $\mathbb{E}(f(X, Y)) = \frac{1}{2\pi} \int_{\mathbb{R}^2} f(x, y) e^{-(x^2+y^2)/2} dx dy = \int_0^{+\infty} f(\sqrt{r} \cos \theta, \sqrt{r} \sin \theta) \frac{1}{2} e^{-r/2} dr \frac{d\theta}{2\pi}$

- using the **rejection** method

Find a probability density g and a constant $c \geq 1$ such that $0 \leq f(x) \leq cg(x)$. Generate i.i.d. variables

$(X^n, U^n) \sim g(x) dx \otimes \mathcal{U}[0, 1]$, compute $r^n = \frac{f(X^n)}{cg(X^n)}$, and accept X^n if $r^n \geq U^n$

SDEs: numerics (2)

- Trajectorial averages: **estimator** $\Phi_{N_{\text{iter}}} = \frac{1}{N_{\text{iter}}} \sum_{n=1}^{N_{\text{iter}}} \varphi(X^n)$
- Numerical scheme ergodic for the probability measure $\psi_{\infty, \Delta t}$
- Two types of errors to compute **averages w.r.t. invariant measure**
 - **Statistical** error, quantified using a Central Limit Theorem

$$\Phi_{N_{\text{iter}}} = \int \varphi \psi_{\infty, \Delta t} + \frac{\sigma_{\Delta t, \varphi}}{\sqrt{N_{\text{iter}}}} \mathcal{G}_{N_{\text{iter}}}, \quad \mathcal{G}_{N_{\text{iter}}} \sim \mathcal{N}(0, 1)$$

- **Systematic** errors
 - **perfect sampling bias**, related to the finiteness of Δt

$$\left| \int_{\mathcal{X}} \varphi \psi_{\infty, \Delta t} - \int_{\mathcal{X}} \varphi \psi_{\infty} \right| \leq C_{\varphi} \Delta t^p$$

- finite sampling bias, related to the finiteness of N_{iter}

SDEs: numerics (3)

Expression of the **asymptotic variance**: correlations matter!

$$\sigma_{\Delta t, \varphi}^2 = \text{Var}(\varphi) + 2 \sum_{n=1}^{+\infty} \mathbb{E}(\tilde{\varphi}(X^n) \tilde{\varphi}(X^0)), \quad \tilde{\varphi} = \varphi - \int \varphi \psi_{\infty, \Delta t}$$

where $\text{Var}(\varphi) = \int_{\mathcal{X}} \tilde{\varphi}^2 \psi_{\infty, \Delta t} = \int_{\mathcal{X}} \varphi^2 \psi_{\infty, \Delta t} - \left(\int_{\mathcal{X}} \varphi \psi_{\infty, \Delta t} \right)^2$

Proof: compute $N_{\text{iter}} \mathbb{E}(\tilde{\Phi}_{N_{\text{iter}}}^2) = \frac{1}{N_{\text{iter}}} \sum_{n,m=0}^{N_{\text{iter}}} \mathbb{E}(\tilde{\varphi}(X^n) \tilde{\varphi}(X^m))$

Stationarity $\mathbb{E}(\tilde{\varphi}(X^n) \tilde{\varphi}(X^m)) = \mathbb{E}(\tilde{\varphi}(X^{n-m}) \tilde{\varphi}(X^0))$ implies

$$N_{\text{iter}} \mathbb{E}(\tilde{\Phi}_{N_{\text{iter}}}^2) = \mathbb{E}(\tilde{\varphi}(X^0)^2) + 2 \sum_{n=1}^{+\infty} \left(1 - \frac{n}{N_{\text{iter}}}\right) \mathbb{E}(\tilde{\varphi}(X^n) \tilde{\varphi}(X^0))$$

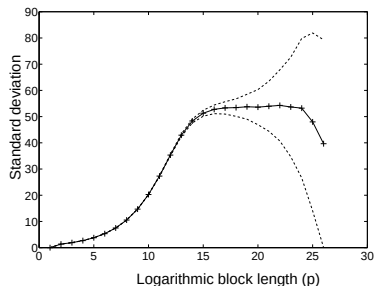
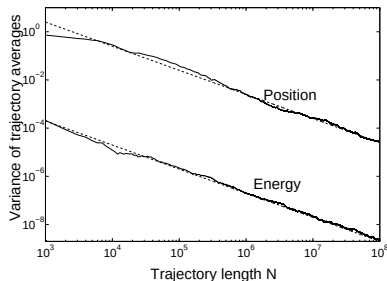
- Useful rewriting: number of **correlated** steps $\sigma_{\Delta t, \varphi}^2 = N_{\text{corr}} \text{Var}(\varphi)$
- Note also that $\sigma_{\Delta t, \varphi}^2 \sim \frac{2}{\Delta t} \mathbb{E} \left[\int_0^{+\infty} \tilde{\varphi}(X_t) \tilde{\varphi}(X_0) dt \right]$

SDEs: numerics (4)

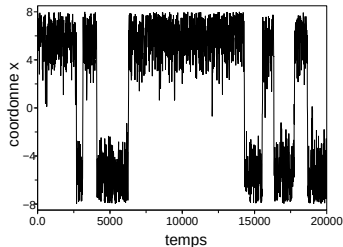
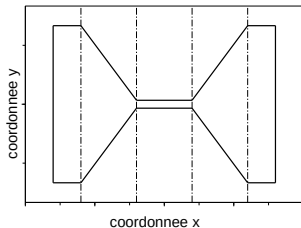
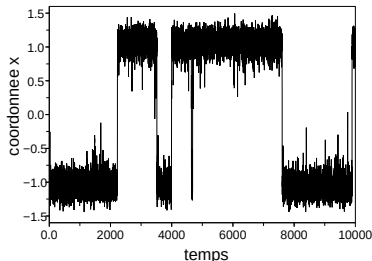
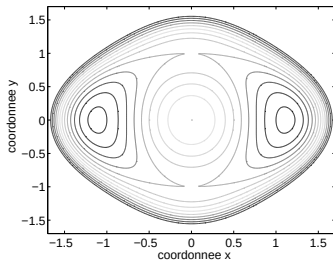
- Estimation of $\sigma_{\Delta t, \varphi}$ by **block averaging** (batch means)

$$\sigma_{\Delta t, \varphi}^2 = \lim_{N, M \rightarrow +\infty} \frac{N}{M} \sum_{k=1}^M \left(\Phi_N^k - \Phi_{NM}^1 \right)^2, \quad \Phi_N^k = \frac{1}{N} \sum_{i=(k-1)N+1}^{kN} \varphi(q^i, p^i)$$

Expected $\Phi_N^k \sim \int_{\mathcal{X}} \varphi \psi_{\infty, \Delta t} + \frac{\sigma_{\Delta t, \varphi}}{\sqrt{N}} \mathcal{G}^k$, with $\mathcal{G}^k$ i.i.d.



Metastability: large variances...



Need for **variance reduction** techniques! (more on Friday)

- **Stochastic differential equations**
 - General perspective (convergence results, ...)
 - Overdamped Langevin dynamics (Einstein-Schmolukowski)
 - Langevin dynamics
 - Extensions: DPD, Generalized Langevin
- **Markov chain methods**
 - Metropolis-Hastings algorithm
- **Deterministic methods**
 - Nosé-Hoover and the like
 - Nosé-Hoover Langevin

Overdamped Langevin dynamics

- SDE on the **configurational** part only (momenta trivial to sample)

$$dq_t = -\nabla V(q_t) dt + \sqrt{\frac{2}{\beta}} dW_t$$

- **Invariance of the canonical measure** $\nu(dq) = \psi_0(q) dq$

$$\psi_0(q) = Z^{-1} e^{-\beta V(q)}, \quad Z = \int_{\mathcal{D}} e^{-\beta V(q)} dq$$

- Generator $\mathcal{L} = -\nabla V(q) \cdot \nabla_q + \frac{1}{\beta} \Delta_q$
 - **invariance** of ψ_0 : adjoint $\mathcal{L}^* \varphi = \operatorname{div}_q \left((\nabla V) \varphi + \frac{1}{\beta} \nabla_q \varphi \right)$
 - elliptic generator hence irreducibility and **ergodicity**
- Discretization $q^{n+1} = q^n - \Delta t \nabla V(q^n) + \sqrt{\frac{2\Delta t}{\beta}} G^n$ (+ **Metropolization**)

Langevin dynamics (1)

- **Stochastic** perturbation of the Hamiltonian dynamics

$$\begin{cases} dq_t = M^{-1} p_t dt \\ dp_t = -\nabla V(q_t) dt - \gamma M^{-1} p_t dt + \sigma dW_t \end{cases}$$

- γ, σ may be matrices, and may depend on q
- **Generator** $\mathcal{L} = \mathcal{L}_{\text{ham}} + \mathcal{L}_{\text{thm}}$

$$\mathcal{L}_{\text{ham}} = p^T M^{-1} \nabla_q - \nabla V(q)^T \nabla_p = \sum_{i=1}^{dN} \frac{p_i}{m_i} \partial_{q_i} - \partial_{q_i} V(q) \partial_{p_i}$$

$$\mathcal{L}_{\text{thm}} = -p^T M^{-1} \gamma^T \nabla_p + \frac{1}{2} (\sigma \sigma^T) : \nabla_p^2 \quad \left(= \frac{\sigma^2}{2} \Delta_p \text{ for scalar } \sigma \right)$$

- **Irreducibility** can be proved (control argument)

Langevin dynamics (2)

- **Invariance** of the canonical measure to conclude to **ergodicity**?

Fluctuation/dissipation relation

$$\sigma \sigma^T = \frac{2}{\beta} \gamma \quad \text{implies} \quad \mathcal{L}^* \left(e^{-\beta H} \right) = 0$$

- Proof for **scalar** γ, σ : a simple computation shows that

$$\mathcal{L}_{\text{ham}}^* = -\mathcal{L}_{\text{ham}}, \quad \mathcal{L}_{\text{ham}} H = 0$$

- Overdamped Langevin analogy $\mathcal{L}_{\text{thm}} = \gamma \left(-p^T M^{-1} \nabla_p + \frac{1}{\beta} \Delta_p \right)$

→ Replace q by p and $\nabla V(q)$ by $M^{-1}p$

$$\mathcal{L}_{\text{thm}}^* \left[\exp \left(-\beta \frac{p^T M^{-1} p}{2} \right) \right] = 0$$

- Conclusion: $\mathcal{L}_{\text{ham}}^*$ and $\mathcal{L}_{\text{thm}}^*$ both preserve $e^{-\beta H(q,p)} dq dp$

Overdamped limit of the Langevin dynamics

- Either $M = \varepsilon \rightarrow 0$ (for $\gamma = 1$) or $\gamma = \frac{1}{\varepsilon} \rightarrow +\infty$ (for $m = 1$ and an appropriate time-rescaling $t \rightarrow t/\varepsilon$)

$$\begin{cases} dq_t^\varepsilon = v_t^\varepsilon dt \\ \varepsilon dv_t^\varepsilon = -\nabla V(q_t^\varepsilon) dt - v_t^\varepsilon dt + \sqrt{\frac{2}{\beta}} dW_t \end{cases}$$

- **Limiting dynamics** $dq_t^0 = -\nabla V(q_t^0) dt + \sqrt{\frac{2}{\beta}} dW_t$
- **Convergence result:** $\lim_{\varepsilon \rightarrow 0} \left(\sup_{0 \leq s \leq t} \|q_s^\varepsilon - q_s^0\| \right) = 0 \text{ (a.s.)}$

The proof relies on the equality

$$\begin{aligned} q_t^\varepsilon - q_t^0 &= v_0 \varepsilon \left(1 - e^{-t/\varepsilon} \right) - \int_0^t \left(1 - e^{-(t-r)/\varepsilon} \right) \left(\nabla V(q_r^\varepsilon) - \nabla V(q_r^0) \right) dr \\ &\quad + \int_0^t e^{-(t-r)/\varepsilon} \nabla V(q_r^0) dr - \sqrt{2} \int_0^t e^{-(t-r)/\varepsilon} dW_r \end{aligned}$$

Numerical integration of the Langevin dynamics (1)

- **Splitting** strategy: Hamiltonian part + fluctuation/dissipation

$$\begin{cases} dq_t = M^{-1} p_t dt \\ dp_t = -\nabla V(q_t) dt \end{cases} \oplus \begin{cases} dq_t = 0 \\ dp_t = -\gamma M^{-1} p_t dt + \sqrt{\frac{2\gamma}{\beta}} dW_t \end{cases}$$

- Hamiltonian part integrated using a Verlet scheme
- **Analytical integration** of the fluctuation/dissipation part

$$d\left(e^{\gamma M^{-1}t} p_t\right) = e^{\gamma M^{-1}t} (dp_t + \gamma M^{-1} p_t dt) = \sqrt{\frac{2\gamma}{\beta}} e^{\gamma M^{-1}t} dW_t$$

so that

$$p_t = e^{-\gamma M^{-1}t} p_0 + \sqrt{\frac{2\gamma}{\beta}} \int_0^t e^{-\gamma M^{-1}(t-s)} dW_s$$

It can be shown that $\int_0^t f(s) dW_s \sim \mathcal{N}\left(0, \int_0^t f(s)^2 ds\right)$

Numerical integration of the Langevin dynamics (2)

- Trotter splitting (define $\alpha_{\Delta t} = e^{-\gamma M^{-1} \Delta t}$, choose $\gamma M^{-1} \Delta t \sim 0.01 - 1$)

$$\left\{ \begin{array}{l} p^{n+1/2} = p^n - \frac{\Delta t}{2} \nabla V(q^n), \\ q^{n+1} = q^n + \Delta t M^{-1} p^{n+1/2}, \\ \tilde{p}^{n+1} = p^{n+1/2} - \frac{\Delta t}{2} \nabla V(q^{n+1}), \\ p^{n+1} = \alpha_{\Delta t} \tilde{p}^{n+1} + \sqrt{\frac{1 - \alpha_{2\Delta t}}{\beta}} M G^n, \end{array} \right.$$

Error estimate on the invariant measure $\mu_{\Delta t}$ of the numerical scheme

There exist a function f such that, for any smooth observable ψ ,

$$\int_{\mathcal{E}} \psi d\mu_{\Delta t} = \int_{\mathcal{E}} \psi d\mu + \Delta t^2 \int_{\mathcal{E}} \psi f d\mu + O(\Delta t^3)$$

- Strang splitting more expensive and not more accurate

Some extensions (1)

- The Langevin dynamics is not Galilean invariant, hence not consistent with **hydrodynamics** → friction forces depending on **relative velocities**

Dissipative Particle Dynamics

$$\begin{cases} dq = M^{-1} p_t dt \\ dp_{i,t} = -\nabla_{q_i} V(q_t) dt + \sum_{i \neq j} \left(-\gamma \chi^2(r_{ij,t}) v_{ij,t} dt + \sqrt{\frac{2\gamma}{\beta}} \chi(r_{ij,t}) dW_{ij} \right) \end{cases}$$

with $\gamma > 0$, $r_{ij} = |q_i - q_j|$, $v_{ij} = \frac{p_i}{m_i} - \frac{p_j}{m_j}$, $\chi \geq 0$, and $W_{ij} = -W_{ji}$

- Invariance of the canonical measure, **preservation** of $\sum_{i=1}^N p_i$
- **Ergodicity** is an issue⁶
- Numerical scheme: splitting strategy⁷

⁶T. Shardlow and Y. Yan, *Stoch. Dynam.* (2006)

⁷T. Shardlow, *SIAM J. Sci. Comput.* (2003)

Some extensions (2)

- **Mori-Zwanzig** derivation⁸ from a generalized Hamiltonian system: particle coupled to **harmonic** oscillators with a **distribution of frequencies**

Generalized Langevin equation ($M = \text{Id}$)

$$\begin{cases} dq = p_t dt \\ dp_t = -\nabla V(q_t) dt + R_t dt \\ \varepsilon dR_t = -R_t dt - \gamma p_t dt + \sqrt{\frac{2\gamma}{\beta}} dW_t \end{cases}$$

- **Invariant measure** $\Pi(q, p, R) = Z_{\gamma, \varepsilon}^{-1} \exp \left(-\beta \left[H(q, p) + \frac{\varepsilon}{2\gamma} R^2 \right] \right)$
- Langevin equation recovered in the limit $\varepsilon \rightarrow 0$
- Ergodicity proofs (hypocoercivity): as for the Langevin equation⁹

⁸R. Kupferman, A. Stuart, J. Terry and P. Tupper, *Stoch. Dyn.* (2002)

⁹M. Ottobre and G. Pavliotis, *Nonlinearity* (2011)

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Metropolis-Hastings algorithm (1)

- Markov chain method^{10,11}, on position space

- Given q^n , propose $\tilde{q}^{n+1}$ according to transition probability $T(q^n, \tilde{q})$
- Accept the proposition with probability $\min(1, r(q^n, \tilde{q}^{n+1}))$ where

$$r(q, q') = \frac{T(q', q) \nu(q')}{T(q, q') \nu(q)}, \quad \nu(dq) \propto e^{-\beta V(q)}.$$

If acceptance, set $q^{n+1} = \tilde{q}^{n+1}$; otherwise, set $q^{n+1} = q^n$.

- Example of proposals

- Gaussian displacement $\tilde{q}^{n+1} = q^n + \sigma G^n$ with $G^n \sim \mathcal{N}(0, \text{Id})$
- Biased random walk^{12,13} $\tilde{q}^{n+1} = q^n - \alpha \nabla V(q^n) + \sqrt{\frac{2\alpha}{\beta}} G^n$

¹⁰Metropolis, Rosenbluth ($\times 2$), Teller ($\times 2$), *J. Chem. Phys.* (1953)

¹¹W. K. Hastings, *Biometrika* (1970)

¹²G. Roberts and R.L. Tweedie, *Bernoulli* (1996)

¹³P.J. Rossky, J.D. Doll and H.L. Friedman, *J. Chem. Phys.* (1978)

Metropolis-Hastings algorithm (2)

- The normalization constant in the canonical measure needs not be known
- **Transition kernel**: accepted moves + rejection

$$P(q, dq') = \min \left(1, r(q, q') \right) T(q, q') dq' + \left(1 - \alpha(q) \right) \delta_q(dq'),$$

where $\alpha(q) \in [0, 1]$ is the probability to accept a move starting from q :

$$\alpha(q) = \int_{\mathcal{D}} \min \left(1, r(q, q') \right) T(q, q') dq'.$$

- The canonical measure is reversible with respect to ν

$$P(q, dq') \nu(dq) = P(q', dq) \nu(dq')$$

This implies **invariance**: $\int_{\mathcal{D}} \psi(q') P(q, dq') \nu(dq) = \int_{\mathcal{D}} \psi(q) \nu(dq)$

Metropolis-Hastings algorithm (3)

- Proof: Detailed balance on the absolutely continuous parts

$$\begin{aligned}\min(1, r(q, q')) T(q, dq') \nu(dq) &= \min(1, r(q', q)) r(q, q') T(q, dq') \nu(dq) \\ &= \min(1, r(q', q)) T(q', dq) \nu(dq')\end{aligned}$$

using successively $\min(1, r) = r \min\left(1, \frac{1}{r}\right)$ and $r(q, q') = \frac{1}{r(q', q)}$

- Equality on the singular parts $(1 - \alpha(q)) \delta_q(dq') \nu(dq) = (1 - \alpha(q')) \delta_{q'}(dq) \nu(dq')$

$$\begin{aligned}\int_{\mathcal{D}} \int_{\mathcal{D}} \phi(q, q') (1 - \alpha(q)) \delta_q(dq') \nu(dq) &= \int_{\mathcal{D}} \phi(q, q) (1 - \alpha(q)) \nu(dq) \\ &= \int_{\mathcal{D}} \int_{\mathcal{D}} \phi(q, q') (1 - \alpha(q')) \delta_{q'}(dq) \nu(dq')\end{aligned}$$

- Note: other acceptance ratios $R(r)$ possible as long as $R(r) = rR(1/r)$, but the Metropolis ratio $R(r) = \min(1, r)$ is optimal in terms of asymptotic variance¹⁴

¹⁴P. Peskun, *Biometrika* (1973)

Metropolis-Hastings algorithm (4)

- **Irreducibility**: for almost all q_0 and any set $\mathcal{S}$ of positive measure, there exists n such that

$$P^n(q_0, \mathcal{S}) = \int_{x \in \mathcal{D}} P(q_0, dx) P^{n-1}(x, \mathcal{S}) > 0$$

- Assume also **aperiodicity** (comes from rejections)

- **Pathwise ergodicity**¹⁵ $\lim_{N_{\text{iter}} \rightarrow +\infty} \frac{1}{N_{\text{iter}}} \sum_{n=1}^{N_{\text{iter}}} A(q^n) = \int_{\mathcal{D}} A(q) \nu(dq)$

- **Central limit theorem** for Markov chains under additional assumptions:

$$\sqrt{N_{\text{iter}}} \left| \frac{1}{N_{\text{iter}}} \sum_{n=1}^{N_{\text{iter}}} A(q^n) - \int_{\mathcal{D}} A(q) \nu(dq) \right| \xrightarrow[N_{\text{iter}} \rightarrow +\infty]{\text{law}} \mathcal{N}(0, \sigma^2)$$

¹⁵S. Meyn and R. Tweedie, *Markov Chains and Stochastic Stability* (1993)

Metropolis-Hastings algorithm (5)

- The asymptotic variance σ^2 takes into account the **correlations**:

$$\sigma^2 = \text{Var}_\nu(A) + 2 \sum_{n=1}^{+\infty} \mathbb{E}_\nu \left[(A(q^0) - \mathbb{E}_\nu(A)) (A(q^n) - \mathbb{E}_\nu(A)) \right]$$

- Numerical efficiency: **trade-off** between acceptance and sufficiently large moves in space to **reduce autocorrelation** (rejection rate around 0.5)¹⁶
- Refined Monte Carlo moves such as
 - “non physical” moves
 - parallel tempering
 - replica exchanges
 - Hybrid Monte-Carlo
- A way to **stabilize discretization schemes for SDEs**

¹⁶Roberts/Gelman/Gilks (1997), ..., Jourdain/Lelièvre/Miasojedow (2012)

- **Stochastic differential equations**
 - General perspective (convergence results, ...)
 - Overdamped Langevin dynamics (Einstein-Schmolukowski)
 - Langevin dynamics
 - Extensions: DPD, Generalized Langevin
- **Markov chain methods**
 - Metropolis-Hastings algorithm
- **Deterministic methods**
 - Nosé-Hoover and the like
 - Nosé-Hoover Langevin

Deterministic methods: Nosé-Hoover and the like (1)

- **Extra variable ξ** mimicking the influence of an energy reservoir
 - friction or anti-friction depending on the sign of ξ
 - “mass” parameter $Q > 0$
 - **feedback mechanism**: increase friction if kinetic temperature too large, decrease otherwise

EDO on extended phase space

$$\begin{cases} \dot{q} = M^{-1}p \\ \dot{p} = -\nabla V(q) - \xi p \\ \dot{\xi} = \frac{1}{Q} \left(p^T M^{-1} p - \frac{3N}{\beta} \right) \end{cases}$$

- Generator $\mathcal{L} = \mathcal{L}_{\text{ham}} + \mathcal{L}_{\text{NH}}$ with $\mathcal{L}_{\text{ham}} = p^T M^{-1} \nabla_q - \nabla V(q)^T \nabla_p$ and

$$\mathcal{L}_{\text{NH}} = -\xi p^T \nabla_p + \frac{1}{Q} \left(p^T M^{-1} p - \frac{3N}{\beta} \right) \partial_{\xi}$$

Deterministic methods: Nosé-Hoover and the like (2)

- A simple computation shows that $\mathcal{L}^* = -\mathcal{L} + 3N\xi$ and

$$\mathcal{L} \left(H(q,p) + \frac{Q\xi^2}{2} \right) = -\frac{3N}{\beta}\xi$$

Invariant measure: solution of $\mathcal{L}^*\pi = 0$

$$\pi(dq dp d\xi) = Z_Q^{-1} e^{-\beta H(q,p)} e^{-\beta Q\xi^2/2} dq dp d\xi$$

Hence (q,p) distributed according to the canonical measure μ

- Discretization: time reversible and measure preserving splitting, or Hamiltonian reformulation
- It converges fast (as $1/N_{\text{iter}}$)... but maybe not to the correct value!
- Ergodicity is an issue!
 - Proofs of non-ergodicity in limiting regimes (KAM tori)¹⁷
 - Practical difficulties when heterogeneities (e.g. very different masses)

¹⁷F. Legoll, M. Luskin and R. Moeckel, *ARMA* (2007), *Nonlinearity* (2009)

Deterministic methods: Nosé-Hoover and the like (3)

- Various (**unsatisfactory**) remedies: Nosé-Hoover **chains**, **massive** Nosé-Hoover thermostating, etc¹⁸
- A more serious remedy: add some **stochasticity**¹⁹
→ Additional **Ornstein-Uhlenbeck process on ξ** , ergodic for $e^{-\beta Q \xi^2/2} d\xi$

Langevin Nosé-Hoover

$$\begin{cases} dq_t = M^{-1} p_t dt \\ dp_t = (-\nabla V(q_t) - \xi_t p_t) dt \\ d\xi_t = \left[Q^{-1} \left(p_t^T M^{-1} p_t - \frac{3N}{\beta} \right) - \gamma \xi_t \right] dt + \sqrt{\frac{2\gamma}{\beta Q}} dW_t \end{cases}$$

Generator $\mathcal{L} = \mathcal{L}_{\text{ham}} + \mathcal{L}_{\text{NH}} + \gamma \mathcal{L}_{\text{thm}}$ with $\mathcal{L}_{\text{thm}} = -\xi \partial_\xi + \frac{1}{\beta Q} \partial_\xi^2$

Dynamics **ergodic** for π

¹⁸M. Tuckerman, *Statistical Mechanics...* (2010)

¹⁹B. Leimkuhler, N. Noorizadeh and F. Theil, *J. Stat. Phys.* (2009)

Sampling other thermodynamic ensembles

Grand canonical ensemble (1)

- Thermodynamic ensembles correspond to constraints satisfied in average
→ constrained maximisation of entropy $S(\rho) = -k_B \int \rho \ln \rho d\lambda$ under conditions $\rho \geq 0$, $\int \rho = 1$ and $\int A_i \rho d\lambda = \mathcal{A}_i$
- Reference measure: uniform number of **indistinguishable** particles

$$\lambda(dq dp dN) = \sum_{n=1}^{+\infty} \frac{1}{n! h^{3N}} 1_{\{(q,p) \in \mathcal{D}^n \times \mathbb{R}^{3n}\}} dq dp \delta_n(dN)$$

- Energy and **particle number fixed in average**, volume exactly fixed
- Lagrange multiplier associated with the number constraint: $\beta\mu$
→ defines the **chemical potential** μ

Grand canonical ensemble (2)

Grand canonical probability measure

$$\rho_{\mu\text{VT}}(dq dp dN) = Z_{\mu\text{VT}}^{-1} \sum_{n=1}^{+\infty} \frac{e^{\beta\mu n}}{n! h^{3N}} e^{-\beta H_n(q,p)} 1_{\{(q,p) \in \mathcal{D}^n \times \mathbb{R}^{3n}\}} dq dp \delta_n(dN)$$

- Numerical method to sample this ensemble
 - **Langevin** trajectories at fixed value of n
(or any other method sampling the canonical ensemble)
 - Variations of particle number using a Metropolis procedure
 - At fixed or random times
 - **Insertion**: choose a location at random and put the particle there
 - **Deletion**: delete a randomly chosen particle
 - Insertion difficult when the system is dense: biasing procedures
- Multispecies extension (chemical equilibria)

Isobaric-isothermal ensemble: fixed geometry (1)

- Consider **isotropic** volume changes: cubic simulation cell $(\mathcal{V}^{1/3}\mathbb{T})^3$
- Reference measure: uniform measure on the volume $\mathcal{V}$

$$\lambda(dq dp d\mathcal{V}) = \mathbf{1}_{\{\mathcal{V}>0\}} \mathbf{1}_{\{q \in (\mathcal{V}^{1/3}\mathbb{T})^{3N}\}} d\mathcal{V} dq dp$$

- Energy and **volume fixed in average**, number of particles exactly fixed
- Lagrange multiplier of the volume constraint: βP (**pressure**)

Isotropic isobaric-isothermal ensemble

$$\mu_{\text{NPT}}(dq dp d\mathcal{V}) = Z_{\text{NPT}}^{-1} e^{-\beta P \mathcal{V}} e^{-\beta H(q,p)} \mathbf{1}_{\{\mathcal{V}>0\}} \mathbf{1}_{\{q \in (\mathcal{V}^{1/3}\mathbb{T})^{3N}\}} dq dp d\mathcal{V}$$

Isobaric-isothermal ensemble: fixed geometry (2)

- First option: combine any technique for canonical sampling + Metropolis moves to change the volume at fixed or random times
- **Dynamical version** à la Langevin or Nosé-Hoover? Introduce...

Extended Hamiltonian

$$\mathcal{H}(q, p, \mathcal{V}, p_{\mathcal{V}}) = \frac{p^T M^{-1} p}{2} + V(q) + P\mathcal{V} + \frac{p_{\mathcal{V}}^2}{2\mathcal{M}}$$

NPT measure = $(q, p, \mathcal{V})$ **marginal** of the associated canonical measure

$$\mu_{\text{NPT}}(dq dp d\mathcal{V}) \propto \left(\int_{\mathbb{R}} e^{-\beta \mathcal{H}(q, p, \mathcal{V}, p_{\mathcal{V}})} dp_{\mathcal{V}} \right) dq dp d\mathcal{V}$$

→ Find a method to sample $e^{-\beta \mathcal{H}(q, p, \mathcal{V}, p_{\mathcal{V}})} dq dp d\mathcal{V} dp_{\mathcal{V}}$

Isobaric-isothermal ensemble: fixed geometry (3)

- Convenient to introduce **scaled variables** $\bar{q} = \frac{q}{\mathcal{V}^{1/3}}$ and $\bar{p} = \mathcal{V}^{1/3}p$
- Motivation for $\bar{p}$? Measure preserving transformation: $dq dp = d\bar{q} d\bar{p}$

Extended Hamiltonian in scaled variables

$$\overline{\mathcal{H}}(\bar{q}, \bar{p}, \mathcal{V}, p_{\mathcal{V}}) = \frac{\bar{p}^T M^{-1} \bar{p}}{2\mathcal{V}^{2/3}} + V\left(\mathcal{V}^{1/3} \bar{q}\right) + P\mathcal{V} + \frac{p_{\mathcal{V}}^2}{2\mathcal{M}}$$

- Associated Hamiltonian dynamics
$$\left\{ \begin{array}{l} d\bar{q}_t = \frac{M^{-1} \bar{p}_t}{\mathcal{V}_t^{2/3}} dt \\ d\bar{p}_t = -\nabla V\left(\mathcal{V}_t^{1/3} \bar{q}_t\right) \mathcal{V}_t^{1/3} dt \\ d\mathcal{V}_t = \frac{p_{\mathcal{V},t}}{\mathcal{M}} dt \\ dp_{\mathcal{V},t} = \left(\overline{\mathcal{P}}(\bar{q}_t, \bar{p}_t, \mathcal{V}_t) - P\right) dt \end{array} \right.$$

$$\text{where } \overline{\mathcal{P}}(\bar{q}, \bar{p}, \mathcal{V}) = \frac{1}{3\mathcal{V}} \left[\frac{\bar{p}^T M^{-1} \bar{p}}{\mathcal{V}^{2/3}} - \nabla V\left(\mathcal{V}^{1/3} \bar{q}\right) \cdot \left(\mathcal{V}^{1/3} \bar{q}\right) \right]$$

Isobaric-isothermal ensemble: fixed geometry (4)

- It is possible to work with scaled variables, but it is sometimes preferable to work with (q, p)

$$dq_t = \mathcal{V}_t^{1/3} d\bar{q}_t + \frac{d\mathcal{V}_t}{3\mathcal{V}_t^{2/3}} \bar{q}_t, \quad dp_t = \mathcal{V}_t^{-1/3} d\bar{p}_t - \frac{d\mathcal{V}_t}{3\mathcal{V}_t^{4/3}} \bar{p}_t$$

- The associated **Hamiltonian dynamics** is

$$\left\{ \begin{array}{l} dq_t = \left(M^{-1} p_t + \frac{p_{\mathcal{V},t}}{3\mathcal{M}\mathcal{V}_t} q_t \right) dt \\ dp_t = \left(-\nabla V(q_t) - \frac{p_{\mathcal{V},t}}{3\mathcal{M}\mathcal{V}_t} p_t \right) dt \\ d\mathcal{V}_t = \frac{p_{\mathcal{V},t}}{\mathcal{M}} dt \\ dp_{\mathcal{V},t} = \left(\mathcal{P}(q_t, p_t, \mathcal{V}_t) - P \right) dt \end{array} \right.$$

with **standard pressure** estimator $\mathcal{P}(q, p, \mathcal{V}) = \frac{1}{3\mathcal{V}} \left[p^T M^{-1} p - q \cdot \nabla V(q) \right]$

Isobaric-isothermal ensemble: fixed geometry (5)

- Associated **generator** $\mathcal{L} = \mathcal{L}_{\text{ham}} + \mathcal{L}_{\text{vol}} + \mathcal{L}_{\text{coupl}}$

$$\mathcal{L}_{\text{vol}} = \frac{p_{\mathcal{V}}}{\mathcal{M}} \partial_{\mathcal{V}} + (\mathcal{P} - P) \partial_{p_{\mathcal{V}}}, \quad \mathcal{L}_{\text{coupl}} = \frac{p_{\mathcal{V}}}{3\mathcal{M}\mathcal{V}} (q^T \nabla_q - p^T \nabla_p)$$

- Simple computations show that $\mathcal{L}^* = -\mathcal{L}$ and that $\mathcal{L}\mathcal{H} = 0$, so

$$\mathcal{L}^* \left(e^{-\beta \mathcal{H}} \right) = 0$$

- Upon adding **fluctuation/dissipation** on p_t and $p_{\mathcal{V},t}$ as

$$\begin{cases} dp_t = \left(-\nabla V(q_t) - \frac{p_{\mathcal{V},t}}{3\mathcal{M}\mathcal{V}_t} p_t \right) dt - \gamma M^{-1} p_t dt + \sqrt{\frac{2\gamma}{\beta}} dW_t \\ dp_{\mathcal{V},t} = \left(\mathcal{P}(q_t, p_t, \mathcal{V}_t) - P \right) dt - \tilde{\gamma} \frac{p_{\mathcal{V},t}}{\mathcal{M}} dt + \sqrt{\frac{2\tilde{\gamma}}{\beta}} d\tilde{W}_t \end{cases}$$

the corresponding Langevin dynamics is **ergodic** w.r.t. NPT measure

Isobaric-isothermal ensemble: fixed geometry (6)

- One drawback: it is unclear whether $\mathcal{V}_t$ **remains positive**!
- Suggestion:^{20,21} introduce strain rate $\varepsilon = \frac{\dot{\mathcal{V}}}{3\mathcal{V}}$ and $p_\varepsilon = \frac{p\mathcal{V}}{3\mathcal{V}}$

$$\left\{ \begin{array}{l} dq_t = \left(M^{-1} p_t + \frac{p_{\varepsilon,t}}{\mathcal{M}} q_t \right) dt \\ dp_t = \left(-\nabla V(q_t) - \left(1 + \frac{1}{N} \right) \frac{p_{\varepsilon,t}}{\mathcal{M}} p_t \right) dt \\ d\mathcal{V}_t = \frac{3p_{\varepsilon,t}\mathcal{V}_t}{\mathcal{M}} dt \\ dp_{\varepsilon,t} = 3\mathcal{V}_t \left(\mathcal{P}(q_t, p_t, \mathcal{V}_t) - P \right) dt + \frac{1}{N} p_t^T M^{-1} p_t dt \end{array} \right.$$

- Volume now **remains positive**: $\mathcal{V}_t = \mathcal{V}_0 \exp \left(3 \int_0^t \frac{p_{\varepsilon,s}}{\mathcal{M}} ds \right)$

²⁰W. Hoover, *Phys. Rev. A* (1985, 1986)

²¹G. Martyna, D. Tobias and M. Klein, *J. Chem. Phys.* (1994)

Isobaric-isothermal ensemble: fixed geometry (7)

- Dynamics no longer Hamiltonian! Denote by $\mathcal{L}$ its generator

Extended energy

$$\mathcal{H}(q, p, \mathcal{V}, p_\varepsilon) = \frac{p^T M^{-1} p}{2} + V(q) + P\mathcal{V} + \frac{p_\varepsilon^2}{2\mathcal{M}}$$

- **Conserved energy:** $\mathcal{L}\mathcal{H} = 0$
- Measure preservation since $\mathcal{L}^* = -\mathcal{L}$

$$\mathcal{L}^* \left(e^{-\beta \mathcal{H}} \right) = 0$$

- Conclusion: upon **adding fluctuation/dissipation** in p and p_ε , the corresponding Langevin dynamics is **ergodic** w.r.t. NPT measure
- Discretization: **splitting** scheme (since no Hamiltonian structure)

Isobaric-isothermal ensemble: anisotropic changes (1)

- Definition of the reference measure

- Matrix $A = [a, b, c] \in \mathbb{R}^{3 \times 3}$ of unit cell vectors, domain AT^3
- Uniform measure on volumes $\frac{dA}{\det(A)^2}$ with $dA = dA_{11}dA_{12} \dots dA_{33}$

Proof: write $A = \det(A)^{1/3} \bar{A}$ so $D(\mathcal{V}) = \int_{\mathbb{R}^9} \delta_{\det(A) - \mathcal{V}}(dA) = \mathcal{V}^3 \int_{\mathbb{R}^9} \delta_{\mathcal{V}(\det(\bar{A}) - 1)}(d\bar{A}) = \mathcal{V}^2 D(1)$

- Modular invariance: if $m \in \mathbb{Z}^{3 \times 3}$ with $\det(m) = 1$, then A and Am represent the same system

Proof: $Am = [m_{11}a + m_{21}b + m_{31}c, \dots]$ represents cells vectors obtained as integer linear combinations of a, b, c , with volume preserved. Since $m^{-1} \in \mathbb{Z}^{3 \times 3}$, integer linear combinations of base vectors of Am allow to recover a, b, c .

Anisotropic isobaric-isothermal ensemble

$$\mu_{\text{NPT}}(dq dp dA) = Z_{\text{NPT}}^{-1} e^{-\beta(P \det(A) + H(q,p))} \mathbf{1}_{\{q \in (AT^3)^N, \det(A) > 0\}} \frac{dq dp dA}{\det(A)^2}$$

Isobaric-isothermal ensemble: anisotropic changes (2)

- Consider $p_\varepsilon = (p_\varepsilon)_{a,b} \in \mathbb{R}^{3 \times 3}$, recall $\text{Tr}(p_\varepsilon^T p_\varepsilon) = \sum_{a,b=1}^3 (p_\varepsilon)_{a,b}^2$
- Stress tensor $\mathbf{P}_{a,b}(q, p, \mathcal{V}) = \frac{1}{\mathcal{V}} \left(\sum_{i=1}^N \frac{p_{i,a} p_{i,b}}{m_i} - [\partial_{q_{i,a}} V(q)] q_{i,b} \right)$
- Dynamics (particle index $i = 1, \dots, N$)

$$\begin{cases} dq_{i,t} = \left(M^{-1} p_{i,t} + \frac{p_{\varepsilon,t}}{\mathcal{M}} q_{i,t} \right) dt \\ dp_{i,t} = \left(-\nabla_{q_i} V(q_t) - \frac{p_{\varepsilon,t}}{\mathcal{M}} p_{i,t} - \frac{1}{3N} \frac{\text{Tr}(p_{\varepsilon,t})}{\mathcal{M}} p_{i,t} \right) dt \\ dA_t = \frac{p_{\varepsilon,t} A_t}{\mathcal{M}} dt \\ dp_{\varepsilon,t} = \left[\det(A_t) \left(\mathbf{P}(q_t, p_t, \det(A_t)) - P \right) + \frac{1}{3N} (p_t^T M^{-1} p_t) \text{Id}_{3 \times 3} \right] dt \end{cases}$$
- Volume remains positive since $\det(A_t) = \det(A_0) \exp \left(\int_0^t \frac{\text{Tr}(p_{\varepsilon,s})}{\mathcal{M}} ds \right)$

Isobaric-isothermal ensemble: anisotropic changes (3)

- Associated **generator** $\mathcal{L} = \mathcal{L}_{\text{ham}} + \mathcal{L}_{\text{vol}} + \mathcal{L}_{\text{coupl}}$

$$\mathcal{L}_{\text{vol}} = \sum_{a,b=1}^3 \frac{(p_\varepsilon A)_{a,b}}{\mathcal{M}} \partial_{A_{a,b}} + \left[\det(A) (\mathbf{P}_{a,b} - P) + \frac{1}{3N} (p^T M^{-1} p) \right] \partial_{(p_\varepsilon)_{a,b}}$$

$$\mathcal{L}_{\text{coupl}} = \frac{1}{\mathcal{M}} \sum_{i=1}^N \left((p_\varepsilon q_i)^T \nabla_{q_i} - (p_\varepsilon p_i)^T \nabla_{p_i} \right) - \frac{1}{3N} \frac{\text{Tr}(p_\varepsilon)}{\mathcal{M}} p^T \nabla_p$$

Extended energy

$$\mathcal{H}(q, p, A, p_\varepsilon) = \frac{p^T M^{-1} p}{2} + V(q) + P \det(A) + \frac{\text{Tr}(p_\varepsilon^T p_\varepsilon)}{2\mathcal{M}}$$

- Energy preservation** $\frac{d\mathcal{H}}{dt} = \mathcal{L}\mathcal{H} = 0$, **adjoint** $\mathcal{L}^* = -\mathcal{L} - \frac{2\text{Tr}(p_\varepsilon)}{\mathcal{M}}$

Proof: use Jacobi identity $\frac{d}{dt}(\det(B(t))) = \det(B(t)) \text{Tr}(B(t)^{-1} \dot{B}(t))$ so that $\frac{d}{dt}(\det(A_t)) = \frac{\det(A_t)}{\mathcal{M}} \text{Tr}(p_{\varepsilon,t})$

Use also $\text{div}_x(Ax) = \text{Tr}(A)$ to obtain $\mathcal{L}_{\text{coupl}}^* = -\mathcal{L}_{\text{coupl}} + \frac{\text{Tr}(p_\varepsilon)}{\mathcal{M}}$ and $\mathcal{L}_{\text{vol}}^* = -\mathcal{L}_{\text{vol}} - \frac{3\text{Tr}(p_\varepsilon)}{\mathcal{M}}$

Isobaric-isothermal ensemble: anisotropic changes (4)

Invariance of the NPT measure

$$\mathcal{L}^* \left(e^{-\beta \mathcal{H}} \det(A)^{-2} \right) = 0$$

Ergodicity upon adding fluctuation/dissipation in p_ϵ and p_t

- Various extensions/refinements/alternatives^{22,23,24}
 - Nosé-Hoover type dynamics or Monte-Carlo moves
 - **symmetrization of A** to avoid spurious cell rotations
 - impose a **stress** $\sigma \in \mathbb{R}^{3 \times 3}$ rather than a scalar pressure²⁵
 - overdamped limits of the dynamics
 - **molecular constraints**

²²G. Martyna, D. Tobias and M. Klein, *J. Chem. Phys.* (1994)

²³D. Quigley, *PhD thesis* (2004) and related articles

²⁴M. Tuckerman, *Statistical Mechanics* (2010)

²⁵E. Hernandez, *J. Chem. Phys.* (2001)

Computation of free energy differences

- **Definition of (relative) free energies**
 - Thermodynamic definitions
 - Alchemical transitions vs reaction coordinates
 - Relation to metastability
- **Computational methods: based on...**
 - simple sampling methods (histogram methods, free energy perturbation)
 - constrained dynamics (thermodynamic integration)
 - nonequilibrium dynamics (Jarzynski equality)
 - adaptive biasing techniques (adaptive biasing force, Wang-Landau, ...)

What is free energy?

- A quantity of physical/chemical interest

Absolute free energy

$$F = -\frac{1}{\beta} \ln Z, \quad Z = \int_{\mathcal{E}} e^{-\beta H(q,p)} dq dp$$

- Motivation (Gibbs, 1902): **Analogy** with macroscopic thermodynamics

$$F = U - TS$$

energy $U = \int_{\mathcal{E}} H \psi$, **entropy** $S = -k_B \int_{\mathcal{E}} \psi \ln \psi$ with $\psi = Z^{-1} e^{-\beta H}$

- Can be analytically computed for ideal gases ($V = 0$), and solids at low temperature
- Usually only **free energy differences** matter! (relative likelihood)

Free energy differences: The alchemical case (1)

- Alchemical transition: indexed by an **external parameter** λ (force field parameter, magnetic field,...)

Alchemical free energy difference

$$F(1) - F(0) = -\beta^{-1} \ln \left(\frac{\int_{\mathcal{E}} e^{-\beta H_1(q,p)} dq dp}{\int_{\mathcal{E}} e^{-\beta H_0(q,p)} dq dp} \right)$$

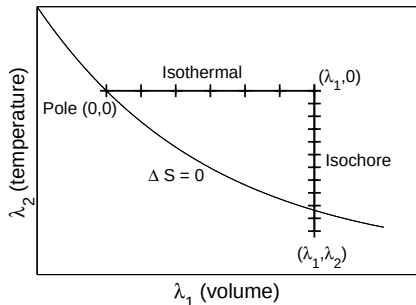
- Typically, $H_\lambda = (1 - \lambda)H_0 + \lambda H_1$
- Example: **Widom insertion** $\rightarrow$ chemical potential $\mu = F(1) - F(0)$

$$V_\lambda(q) = \sum_{1 \leq i < j \leq N} v(|q^i - q^j|) + \lambda \sum_{1 \leq i \leq N} v(|q^i - q^{N+1}|)$$

Free energy differences: The alchemical case (2)

- Another example: Computation of **entropy variation** in a release wave

$$S(\lambda_1, \lambda_2) - S(0, 0) = \frac{1}{k_B T(\lambda_2)} \langle H \rangle_{V(\lambda_1), T(\lambda_2)} - \frac{1}{k_B T(0)} \langle H \rangle_{V(0), T(0)} \\ + \ln \left(\frac{\int_{\Omega(\lambda_1)} e^{-H(q,p)/k_B T(\lambda_2)} dq dp}{\int_{\Omega(0)} e^{-H(q,p)/k_B T(0)} dq dp} \right)$$



Free energy differences: The reaction coordinate case (1)

- **Reaction coordinate** $\xi : \mathcal{D} \rightarrow \mathbb{R}^m$ (angle, length, ...)
- **Foliation** of the configurational space using level sets of ξ

$$\mathcal{D} = \bigcup_{z \in \mathbb{R}^m} \Sigma(z), \quad \Sigma(z) = \left\{ q \in \mathcal{D} \mid \xi(q) = z \right\}$$

Free energy difference: relative likelihood of **marginals in ξ**

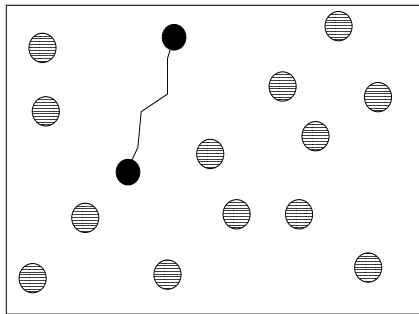
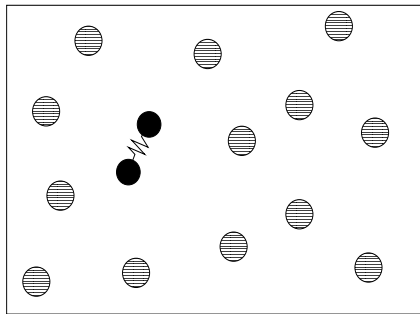
$$F(z_1) - F(z_0) = -\beta^{-1} \ln \left(\frac{\int_{\Sigma(z_1) \times \mathbb{R}^{3N}} e^{-\beta H(q,p)} \delta_{\xi(q)-z_1}(dq) dp}{\int_{\Sigma(z_0) \times \mathbb{R}^{3N}} e^{-\beta H(q,p)} \delta_{\xi(q)-z_0}(dq) dp} \right).$$

with (as in the microcanonical case) $\delta_{\xi(q)-z}(dq) = \frac{\sigma_{\Sigma(z)}(dq)}{|\nabla \xi(q)|}$

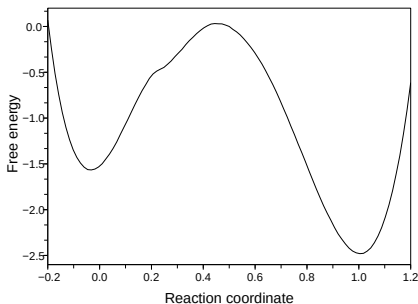
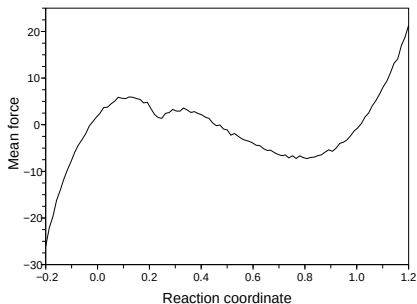
- Depends on the **choice of ξ** and not only on the foliation

Free energy differences: The reaction coordinate case (2)

- Two particles (q_1, q_2), interaction $V_S(r) = h \left[1 - \frac{(r - r_0 - w)^2}{w^2} \right]^2$
- Solvent: purely repulsive potential $V_{\text{WCA}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] + \varepsilon$ if $r \leq r_0$, and 0 for $r > r_0$
- Choose $\xi(q) = \frac{|q_1 - q_2| - r_0}{2w}$ (0 for compact, 1 for stretched)



Free energy differences: The reaction coordinate case (3)



Left: Estimated mean force $F'(z)$.

Right: Corresponding potential of mean force $F(z)$.

Parameters: $\beta = 1$, $N = 100$ particles, solvent density $\rho = 0.436$, WCA interactions $\sigma = 1$ and $\varepsilon = 1$, dimer $w = 2$ and $h = 2$.

Free energy based importance sampling (1)

- In many cases, the probability distribution μ to sample is multimodal
→ **metastability**
- Limit the metastability using **importance sampling**: sample a modified distribution $\tilde{\mu}$

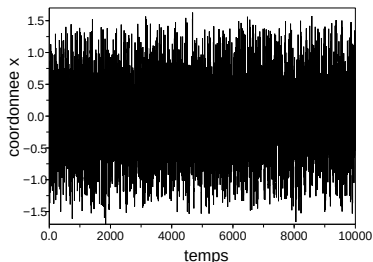
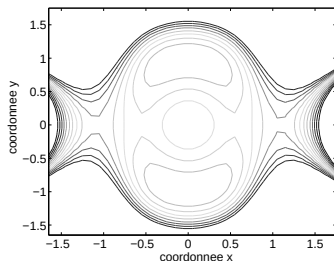
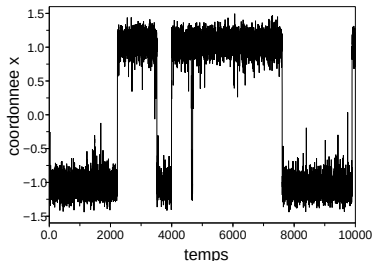
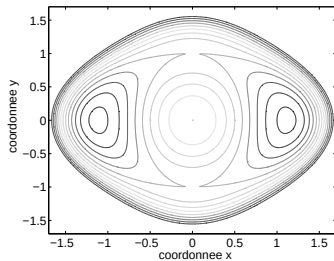
$$\tilde{\mu}(dq dp) = e^{\beta W(q)} \mu(dq dp)$$

and **correct averages** as

$$\mathbb{E}_{\mu}(A) = \frac{\mathbb{E}_{\tilde{\mu}}\left(A e^{-\beta W}\right)}{\mathbb{E}_{\tilde{\mu}}\left(e^{-\beta W}\right)} = \lim_{T \rightarrow +\infty} \frac{\int_0^T A(q_t, p_t) e^{-\beta W(q_t)} dt}{\int_0^T e^{-\beta W(q_t)} dt}$$

- For the canonical measure, $\tilde{\mu}$ is sampled by a Langevin dynamics with **modified potential** $V - W$
- Use $W(q) = F(\xi(q))$ where ξ describe the metastable direction

Free energy based importance sampling (2)



- **Definition of (relative) free energies**
 - Thermodynamic definitions
 - Alchemical transitions vs reaction coordinates
 - Relation to metastability
- **Computational methods: based on...**
 - simple sampling methods (histogram methods, free energy perturbation)
 - constrained dynamics (thermodynamic integration)
 - nonequilibrium dynamics (Jarzynski equality)
 - adaptive biasing techniques (adaptive biasing force, Wang-Landau, ...)

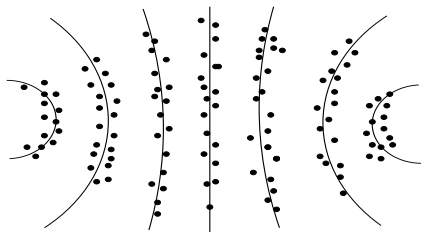
Classification of available methods

- Increasing order of mathematical complexity

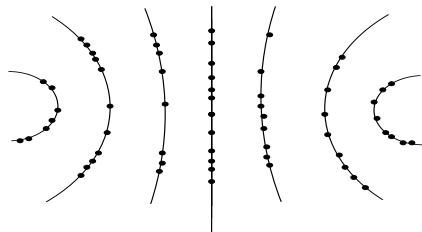
Free energy perturbation	→	Homogeneous MCs and SDEs
Histogram methods	→	Homogeneous MCs and SDEs
Thermodynamic integration	→	Projected MCs and SDEs
Nonequilibrium dynamics	→	Nonhomogenous MCs and SDEs
Adaptive dynamics	→	Nonlinear SDEs and MCs

- On top of that: selection procedures can be added → particle systems and jump processes
- Questions:
 - Consistency (convergence)
 - Efficiency (error estimates = rate of convergence)

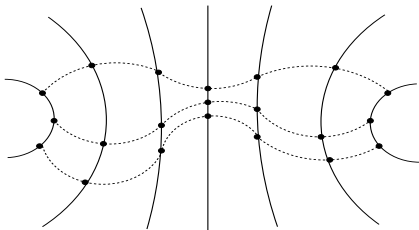
A cartoon comparison of available methods



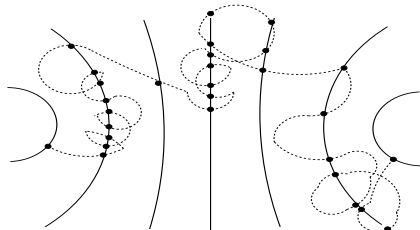
(a) Histogram methods



(b) Thermodynamic integration



(c) Nonequilibrium dynamics



(d) Adaptive dynamics

Free energy perturbation (1)

- **Alchemical** case only! Express ΔF as an average²⁶

$$F(\lambda) - F(0) = -\beta^{-1} \ln \frac{\int_{\mathcal{E}} e^{-\beta(H_{\lambda}(q,p) - H_0(q,p))} \mu_0(dq dp)}{\int_{\mathcal{E}} \mu_0(dq dp)}$$

with $\mu_0(dq dp) = Z^{-1} e^{-\beta H_0(q,p)} dq dp$

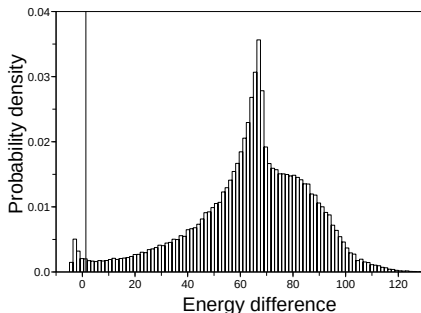
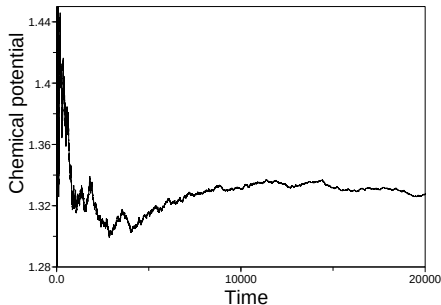
- All **usual sampling techniques** can be used to sample from μ_0
- Simplest estimator

$$\widehat{\Delta F}_M = -\frac{1}{\beta} \ln \left(\frac{1}{M} \sum_{i=1}^M e^{-\beta(H_1(q^i, p^i) - H_0(q^i, p^i))} \right), \quad (q^i, p^i) \sim \mu_0$$

²⁶Zwanzig, *J. Chem. Phys.* **22**, 1420 (1954)

Free energy perturbation (2)

Widom insertion. Left: Estimate of the chemical potential. Right: Distribution $P_0(dU)$ of insertion energies $U = H_1 - H_0$.



- The convergence is plagued by a very **large variance**... Remedies?

- **Staging** (**stratification**):
$$F(1) - F(0) = \sum_{i=1}^I F(\lambda_{i+1}) - F(\lambda_i)$$

Free energy perturbation (3)

- Umbrella sampling²⁷ (importance sampling)

$$F(\lambda) - F(0) = -\beta^{-1} \ln \frac{\int_{\mathcal{E}} e^{-\beta(H_{\lambda}-W)} d\mu_W}{\int_{\mathcal{E}} e^{-\beta(H_0-W)} d\mu_W}, \quad \mu_W \propto \mu_0 e^{-\beta W}$$

- Bridge sampling²⁸: sample from the two distributions μ_0, μ_1 and optimize α to reduce the (asymptotic) variance

$$\frac{Z_1}{Z_0} = \frac{\int_{\mathcal{E}} \alpha e^{-\beta H_1} d\mu_0}{\int_{\mathcal{E}} \alpha e^{-\beta H_0} d\mu_1}, \quad \hat{r}^{n_1, n_2} = \frac{\frac{1}{n_2} \sum_{j=1}^{n_2} \frac{f_1(x^{2,j})}{n_1 f_1(x^{2,j}) + n_2 \hat{r}^{n_1, n_2} f_2(x^{2,j})}}{\frac{1}{n_1} \sum_{j=1}^{n_1} \frac{f_2(x^{1,j})}{n_1 f_1(x^{1,j}) + n_2 \hat{r}^{n_1, n_2} f_2(x^{1,j})}}$$

²⁷G.M. Torrie and J.P. Valleau, *J. Comp. Phys.* **23**, 187 (1977)

²⁸C. Bennett, *J. Comput. Phys.* **22**, pp. 245–268 (1976)

Thermodynamic integration: Alchemical case

- Free energy = **integral of an average force**²⁹

$$F(1) - F(0) = \int_0^1 F'(\lambda) d\lambda \simeq \sum_{i=1}^M (\lambda_i - \lambda_{i-1}) F'(\lambda_i)$$

- Average force**: computed by any method sampling the canonical measure

$$F'(\lambda) = \mathbb{E}_{\mu_\lambda} \left(\frac{\partial H_\lambda}{\partial \lambda} \right), \quad \mu_\lambda(dq dp) = Z_\lambda^{-1} e^{-\beta H_\lambda(q,p)} dq dp$$

- Optimization of the quadrature points to minimize the variance
- Extension to the case of reaction coordinates using **projected SDEs**,
mean force = **average Lagrange multiplier** of the constraint³⁰

²⁹Kirkwood, *J. Chem. Phys.* **3**, 300 (1935)

³⁰Ciccotti, Lelièvre, Vanden-Eijnden, *Comm. Pure Appl. Math.* (2008)

Thermodynamic integration: Constrained Langevin (1)

Constrained Langevin dynamics

$$\begin{cases} dq_t = M^{-1} p_t dt, \\ dp_t = -\nabla V(q_t) dt - \gamma(q_t) M^{-1} p_t dt + \sigma(q_t) dW_t + \nabla \xi(q_t) d\lambda_t, \\ \xi(q_t) = z \end{cases}$$

- Standard fluctuation/dissipation relation $\sigma \sigma^T = \frac{2}{\beta} \gamma$
- **Hidden velocity constraint:** $\frac{d\xi(q_t)}{dt} = v_\xi(q_t, p_t) = \nabla \xi(q_t)^T M^{-1} p_t = 0$
- The corresponding phase-space is $\Sigma_{\xi, v_\xi}(z, 0)$ where

$$\Sigma_{\xi, v_\xi}(z, v_z) = \left\{ (q, p) \in \mathbb{R}^{6N} \mid \xi(q) = z, v_\xi(q, p) = v_z \right\}$$

- An **explicit expression of the Lagrange multiplier** can be found by computing the second derivative in time of the constraint

Thermodynamic integration: Constrained Langevin (2)

Invariant measure

$$\mu_{\Sigma_{\xi, v_{\xi}}(z, 0)}(dq dp) = Z_{z, 0}^{-1} e^{-\beta H(q, p)} \sigma_{\Sigma_{\xi, v_{\xi}}(z, 0)}(dq dp)$$

with $\sigma_{\Sigma_{\xi, v_{\xi}}(z, v_z)}(dq dp)$ phase space Liouville measure induced by J

- Reversibility and detailed balance up to momentum reversal, **ergodicity**
- The free energy can be estimated from constrained samplings as

$$F(z) = F_{\text{rgd}}^M(z) - \frac{1}{\beta} \ln \int_{\Sigma_{\xi, v_{\xi}}(z, 0)} (\det \nabla \xi^T M^{-1} \nabla \xi)^{-1/2} d\mu_{\Sigma_{\xi, v_{\xi}}(z, 0)} + C$$

with **rigid free energy** $F_{\text{rgd}}^M(z) = -\frac{1}{\beta} \ln \int_{\Sigma_{\xi, v_{\xi}}(z, 0)} e^{-\beta H(q, p)} d\mu_{\Sigma_{\xi, v_{\xi}}(z, 0)}$

- Thermodynamic integration through the computation of the **mean force**

$$\nabla_z F_{\text{rgd}}^M(z) = \int_{\Sigma_{\xi, v_{\xi}}(z, 0)} f_{\text{rgd}}^M(q, p) \mu_{\Sigma_{\xi, v_{\xi}}(z, 0)}(dq dp)$$

Thermodynamic integration: Constrained Langevin (3)

- **Splitting** into Hamiltonian & constrained Ornstein-Uhlenbeck
- Midpoint scheme for momenta (**reversible** for constrained measure)

$$p^{n+1/4} = p^n - \frac{\Delta t}{4} \gamma M^{-1} (p^n + p^{n+1/4}) + \sqrt{\frac{\Delta t}{2}} \sigma G^n + \nabla \xi(q^n) \lambda^{n+1/4},$$

with $\lambda^{n+1/4}$ such that $\nabla \xi(q^n)^T M^{-1} p^{n+1/4} = 0$

- **RATTLE scheme** (symplectic)

$$\begin{cases} p^{n+1/2} &= p^{n+1/4} - \frac{\Delta t}{2} \nabla V(q^n) + \nabla \xi(q^n) \lambda^{n+1/2}, \\ q^{n+1} &= q^n + \Delta t M^{-1} p^{n+1/2}, \\ p^{n+3/4} &= p^{n+1/2} - \frac{\Delta t}{2} \nabla V(q^{n+1}) + \nabla \xi(q^{n+1}) \lambda^{n+3/4}, \end{cases}$$

where $\lambda^{n+1/2}$ and $\lambda^{n+3/4}$ are such that

$$\xi(q^{n+1}) = z, \quad \nabla \xi(q^{n+1})^T M^{-1} p^{n+3/4} = 0$$

Thermodynamic integration: Constrained Langevin (4)

- **Metropolization** of the RATTLE part to eliminate the time-step error in the sampled measure
- Longtime (a.s.) convergence (No second order derivatives of ξ needed)

$$\lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T d\lambda_t = \nabla_z F_{\text{rgd}}^M(z)$$

- **Variance reduction**: keep only the Hamiltonian part of λ_t
- Numerical discretization: only **Lagrange multipliers from RATTLE**:

$$\nabla_z F_{\text{rgd}}^M(z) \simeq \frac{1}{N} \sum_{n=0}^{N-1} f_{\text{rgd}}^M(q^n, p^n) \simeq \frac{1}{N\Delta t} \sum_{n=0}^{N-1} (\lambda^{n+1/2} + \lambda^{n+3/4})$$

- Consistency result

$$\lambda^{n+1/2} + \lambda^{n+3/4} = \frac{\Delta t}{2} \left(f_{\text{rgd}}^M(q^n, p^{n+1/4}) + f_{\text{rgd}}^M(q^{n+1}, p^{n+3/4}) \right) + O(\Delta t^3)$$

Nonequilibrium dynamics (1)

- Basic idea: switch from the initial to the final state in a finite time, **starting from equilibrium**, and reweight trajectories appropriately³¹
- Simplest possible setting: schedule $\Lambda(0) = 0, \Lambda(T) = 1$

$$\begin{cases} \dot{q}(t) = \nabla_p H_{\Lambda(t)}(q(t), p(t)) \\ \dot{p}(t) = -\nabla_q H_{\Lambda(t)}(q(t), p(t)) \end{cases}$$

- **Work** $\mathcal{W}(q, p) = \int_0^T \frac{\partial H_{\Lambda(t)}}{\partial \lambda}(\phi_t^\Lambda(q, p)) \Lambda'(t) dt = H_1(\phi_T^\Lambda(q, p)) - H_0(q, p)$

Jarzynski equality: exponential reweighting of the works

$$\mathbb{E}_{\mu_0} \left(e^{-\beta \mathcal{W}} \right) = Z_0^{-1} \int_{\mathcal{E}} e^{-\beta H_1(\phi_T^\Lambda(q, p))} dq dp = \frac{Z_1}{Z_0} = e^{-\beta(F(1) - F(0))}$$

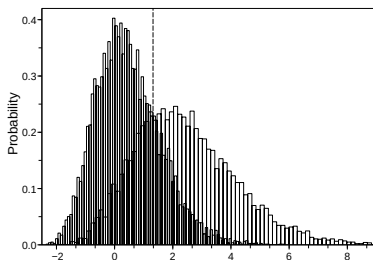
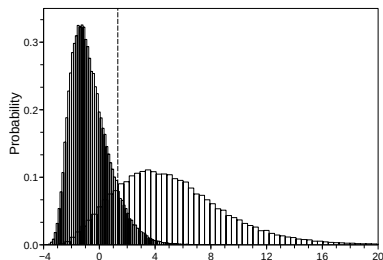
³¹C. Jarzynski, *Phys. Rev. Lett. & Phys. Rev. E* (1997)

Nonequilibrium dynamics (2)

- Mostly of theoretical interest: **weight degeneracies** (same as FEP)
- **Free energy inequality** $\mathbb{E}(\mathcal{W}_t) \geq F(\Lambda(t)) - F(0)$ (Jensen)
- Many extensions...
 - Metropolis dynamics, SDEs (Langevin/overdamped Langevin)
 - Forward/backward versions (Crooks), bridge estimators

$$\frac{Z_T}{Z_0} \mathbb{E} \left(\varphi_{[0,T]}^r(X^b) e^{-\beta \theta \mathcal{W}_{0,T}^b} \right) = \mathbb{E} \left(\varphi_{[0,T]}^f(X^f) e^{-\beta(1-\theta) \mathcal{W}_{0,T}^f} \right)$$

- Driven constrained processes (reaction coordinate case)



Adaptive biasing force (1)

- Simplified setting: $q = (x, y)$ and $\xi(q) = x \in \mathbb{R}$ so that

$$F(x_2) - F(x_1) = -\beta^{-1} \ln \left(\frac{\bar{\nu}(x_2)}{\bar{\nu}(x_1)} \right), \quad \bar{\nu}(x) = \int e^{-\beta V(x,y)} dy$$

- The **mean force** is $F'(x) = \frac{\int \partial_x V(x, y) e^{-\beta V(x,y)} dy}{\int e^{-\beta V(x,y)} dy}$
- The dynamics $dq_t = -\nabla V(q_t) dt + \sqrt{\frac{2}{\beta}} dW_t$ is **metastable**, contrarily to

$$\begin{cases} dq_t = -\nabla \left(V(q_t) - F(\xi(q_t)) \right) dt + \sqrt{\frac{2}{\beta}} dW_t \\ F'(x) = \mathbb{E}_\nu \left(\partial_x V(q) \mid \xi(q) = x \right) \end{cases}$$

- Stationary solution $\psi_\infty \propto e^{-\beta(V - F \circ \xi)}$

Adaptive biasing force (2)

- Note that, for any measure $\tilde{\nu}(dq) \propto \nu(dq) g(\xi(q))$ (with $g \geq 0$)

$$F'(x) = \mathbb{E}_{\tilde{\nu}}\left(\partial_x V(q) \mid \xi(q) = x\right) = \frac{\int \partial_x V(x, y) e^{-\beta V(x, y)} g(x) dy}{\int e^{-\beta V(x, y)} g(x) dy}$$

- Bias the dynamics by an approximation of F' computed **on-the-fly**
→ Replace equilibrium expectations by $F'(t, x) = \mathbb{E}\left(\partial_x V(q_t) \mid \xi(q_t) = x\right)$

ABF dynamics

$$\begin{cases} dq_t = -\nabla\left(V(q_t) - F_t(\xi(q_t))\right) dt + \sqrt{\frac{2}{\beta}} dW_t \\ F'_t(x) = \mathbb{E}\left(\partial_x V(q) \mid \xi(q_t) = x\right) \end{cases}$$

Adaptive biasing force (3)

- Reformulation as a nonlinear PDE on the law $\psi(t, q)$

$$\begin{cases} \partial_t \psi = \operatorname{div} \left[\nabla (V - F_{\text{bias}}(t, x)) \psi + \beta^{-1} \nabla \psi \right], \\ F'_{\text{bias}}(t, x) = \frac{\int \partial_x V(x, y) \psi(t, x, y) dy}{\int \psi(t, x, y) dy}. \end{cases}$$

- Stationary solution is indeed $\psi_\infty \propto e^{-\beta(V - F \circ \xi)}$
- Precise **convergence results** showing that the metastability is removed
- Adaptive methods where the **biasing potential** is updated:
 - Wang-Landau
 - nonequilibrium metadynamics, well-tempered metadynamics, ...
 - Self-healing Umbrella sampling

- **Constrained stochastic dynamics** (equilibrium and nonequilibrium)

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- **Selection mechanisms**

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